

Meningococcal disease: still with us ... but for how long?

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OXFORD

Dedication and disclaimer

This talk is dedicated to all of those who have striven over the past 220 years to reduce the impact of invasive meningococcal disease.

This has been a collective endeavour and the choice of individuals to highlight in this talk is a personal one, it is not, and is certainly not intended to be, in any way comprehensive.

The Maiden Laboratory
NIBSC 1988-1997
Zoology/Biology Oxford 1997-



Meningococcal disease: still with us

Peltola, H. (1983). *Rev Infect Dis* 5, 71-91.



Vieusseux, G. (1806). Mémoire sur la maladie qui a régné à Genève au printemps de 1805. *J Med Chir Pharm* 11, 163-182.

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Hospital neglect a factor in girl's meningitis death

Girl's meningitis leads to quadruple amputation

FAMILY PHOTOGRAPH
Lila Marsland was described as "a lively and vibrant little girl"

Lynette Horsburgh
BBC News, Lancashire

5 June 2025

The death of five-year-old girl who was sent home from hospital suspected tonsillitis hours before dying from meningitis was caused by neglect, an inquest jury has concluded.

FAMILY HANDOUT
Five-year-old Saffanah, from Stockton-on-Tees, is back to her "sassy self", her family said

Pamela Tickell
BBC News, North East and Cumbria

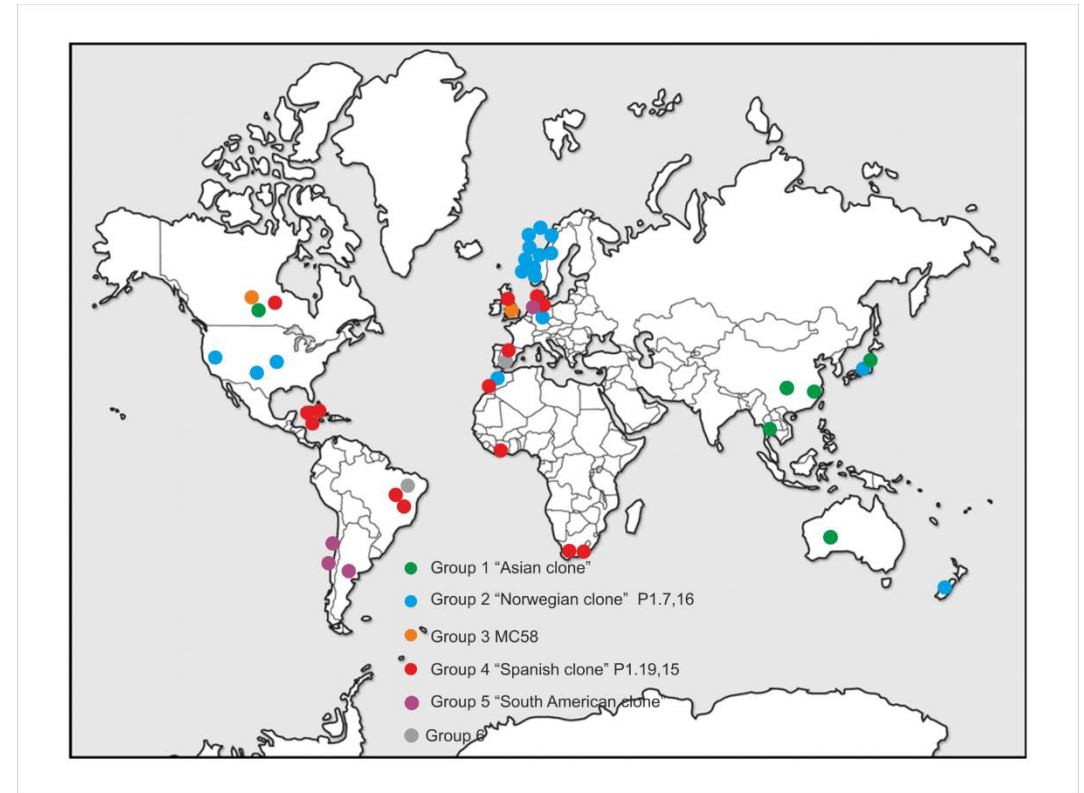
<https://www.bbc.co.uk/news/articles/cx2vv81k2k1o>

April 2025

Global epidemiology of meningococcal disease

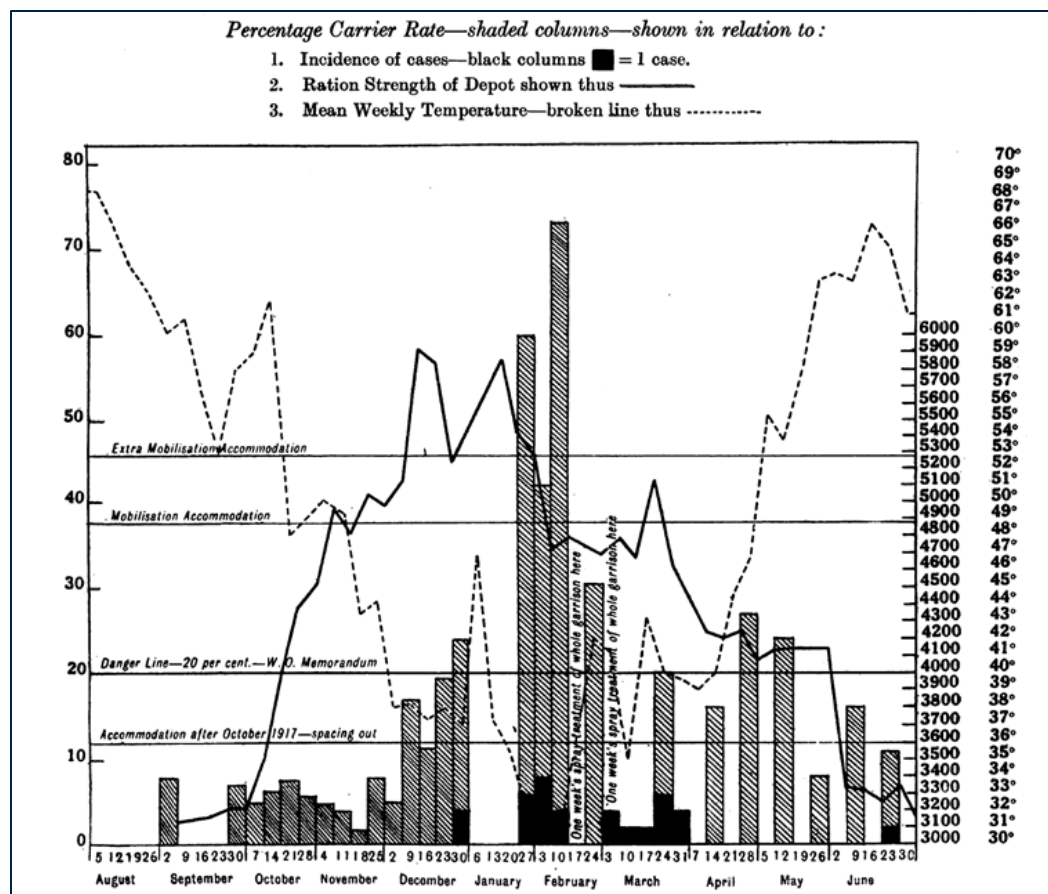
- **Endemic disease:**
 - globally distributed;
 - 0.1 - 5 cases per 100,000.
- **Hyperendemic disease:**
 - Local (community), national, global;
 - ~10 cases per 100,000.
- **Disease outbreaks.**
 - local & institutional (army camp, university).
- **Epidemic/pandemic disease.**
 - Regional, national, & global;
 - 500 cases per 100,000, or more.

Harrison, L. H., Trotter, C. L. & Ramsay, M. E. (2009). Global epidemiology of meningococcal disease. *Vaccine* **27**, B51-B63.

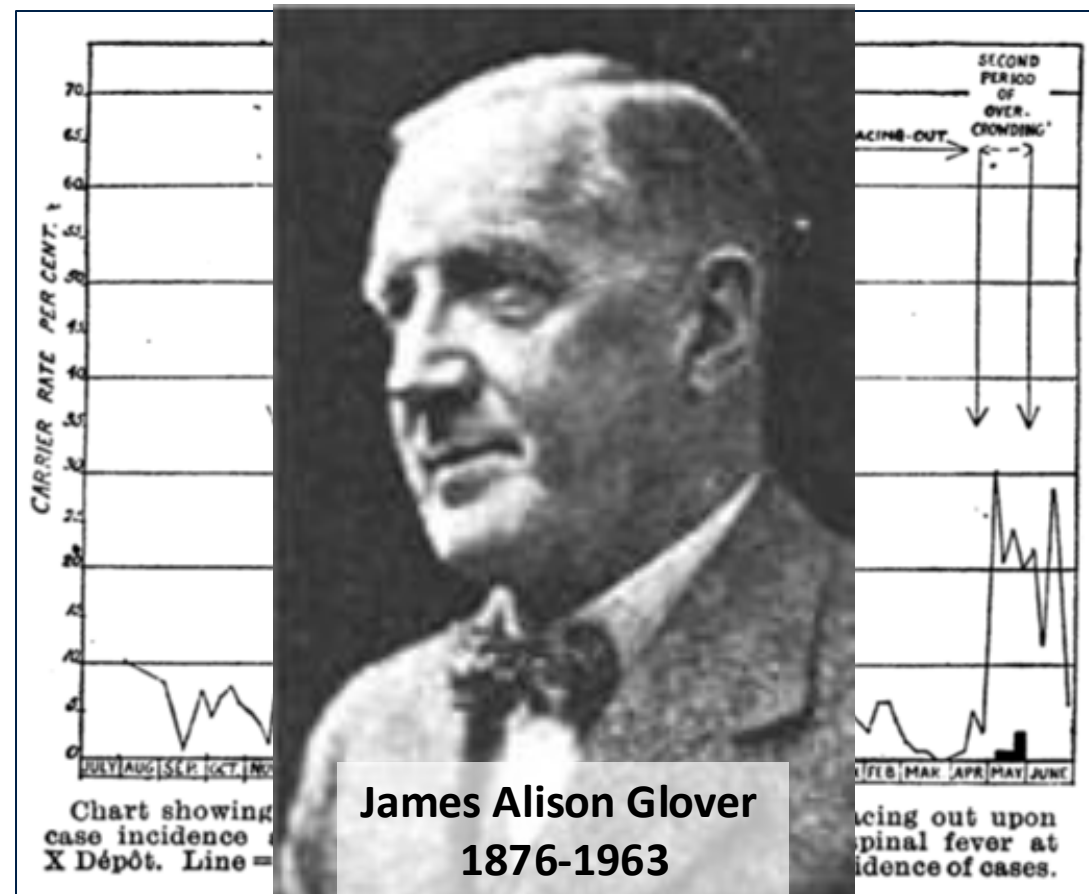


Harrison, O. B., Bray, J. E., Maiden, M. C. & Caugant, D. A. (2015). Genomic Analysis of the Evolution and Global Spread of Hyper-invasive Meningococcal Lineage 5. *EBioMedicine* **2**, 234-243.

1917: Disease, carriage, and 'social distancing'

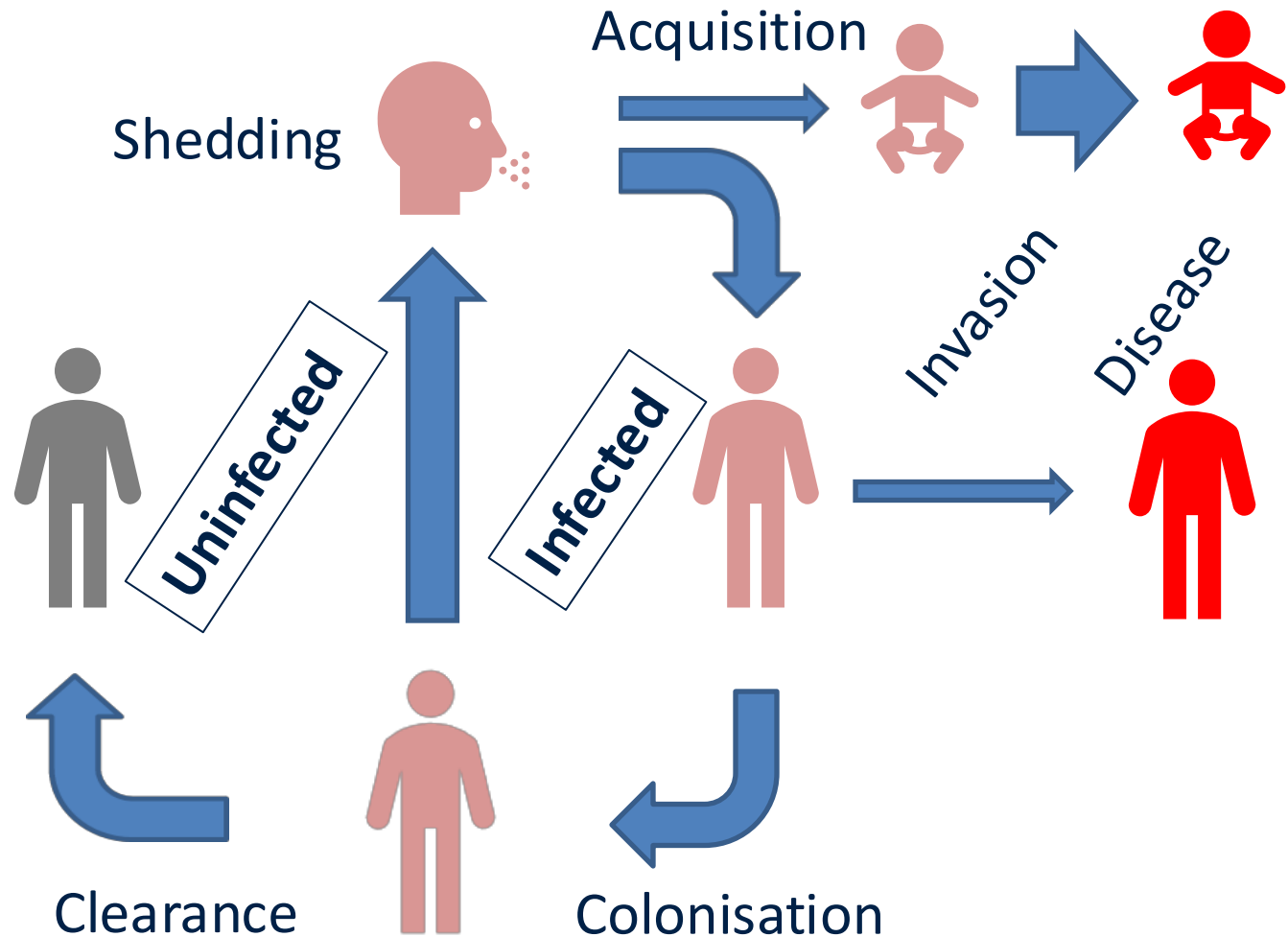


Glover, J. A. (1918). The Cerebro-Spinal Fever Epidemic of 1917 at X Depot. *J Hyg (Lond)* **17**, 350-365.



Glover, J. A. (1918). "Spacing out" in the Prevention of Military Epidemics of Cerebro-Spinal Fever. *Br Med J.* **2**, 509-512.

Meningococcal Transmission, Infection, and Invasion

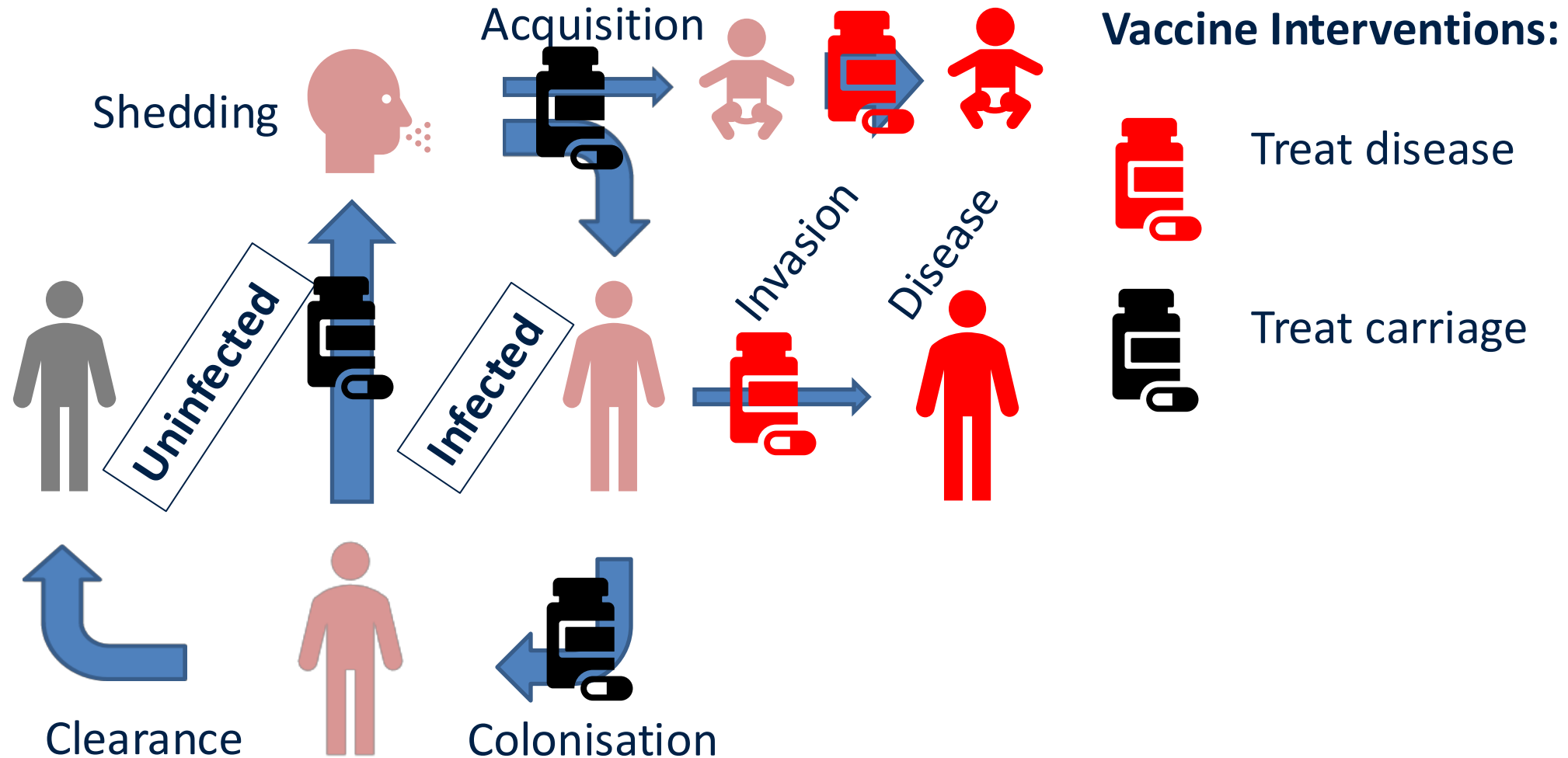


The meningococcus is ordinarily a commensal, causing disease rarely.

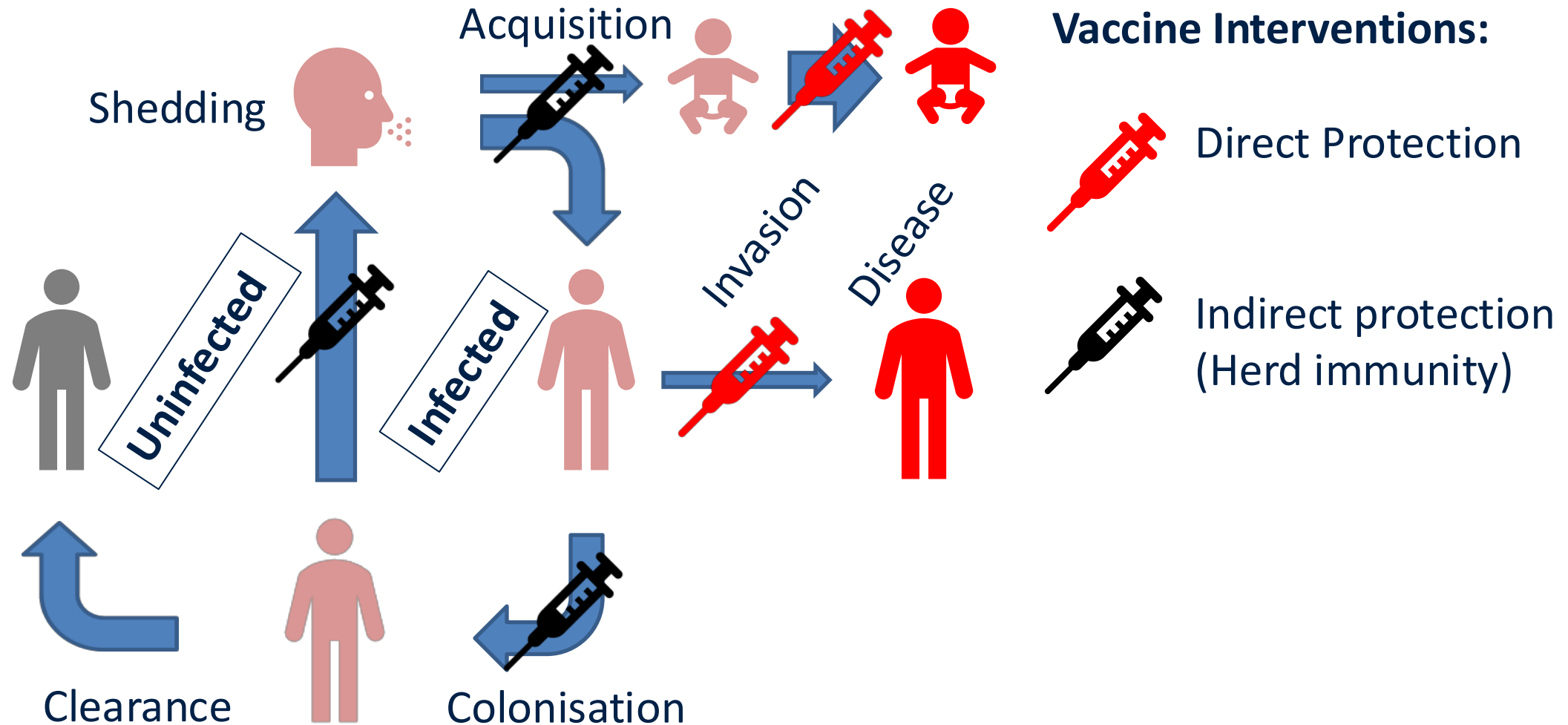
Invasion plays no role in transmission, so it can be thought of as an **accidental pathogen**.

Trotter, C. L. & Maiden, M. C. (2009). Meningococcal vaccines and herd immunity: lessons learned from serogroup C conjugate vaccination programs. *Expert Rev Vaccines* 8, 851-861.

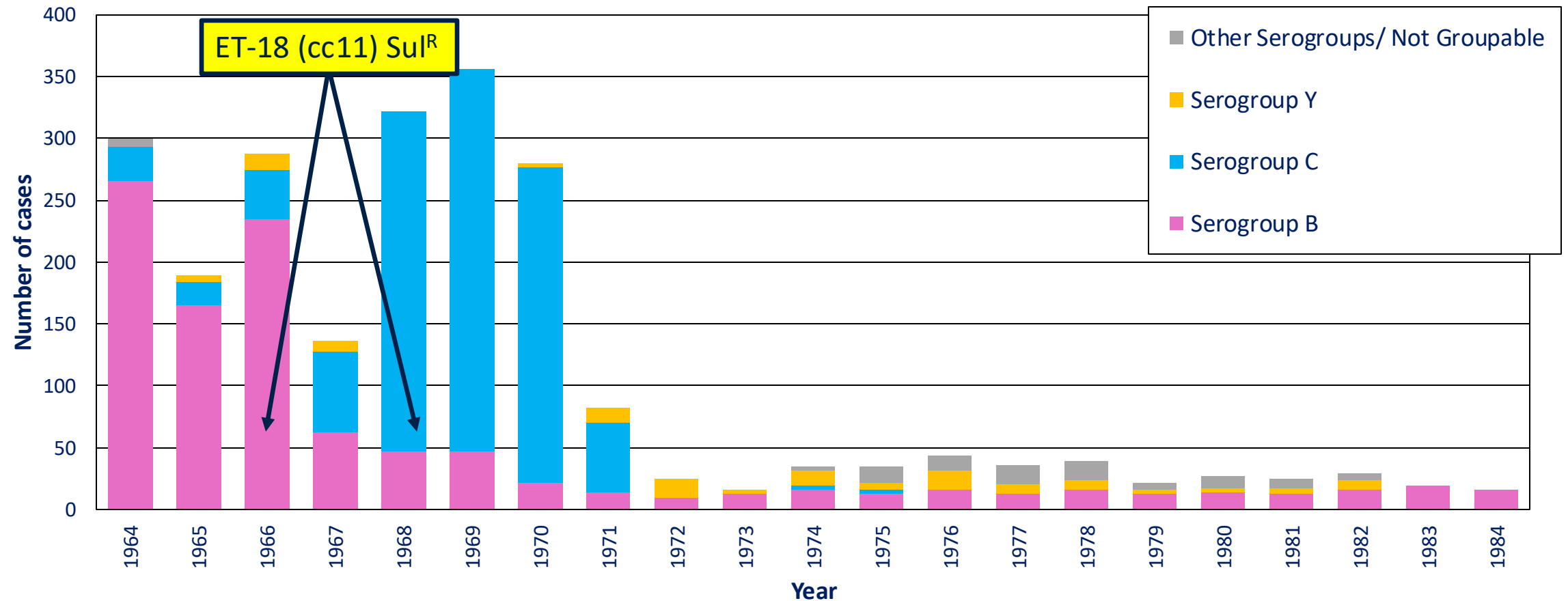
Protecting against meningococcal disease: antimicrobials



Protecting against meningococcal disease: vaccines



USA Army Outbreaks: AMR and the development of plain polysaccharide vaccines

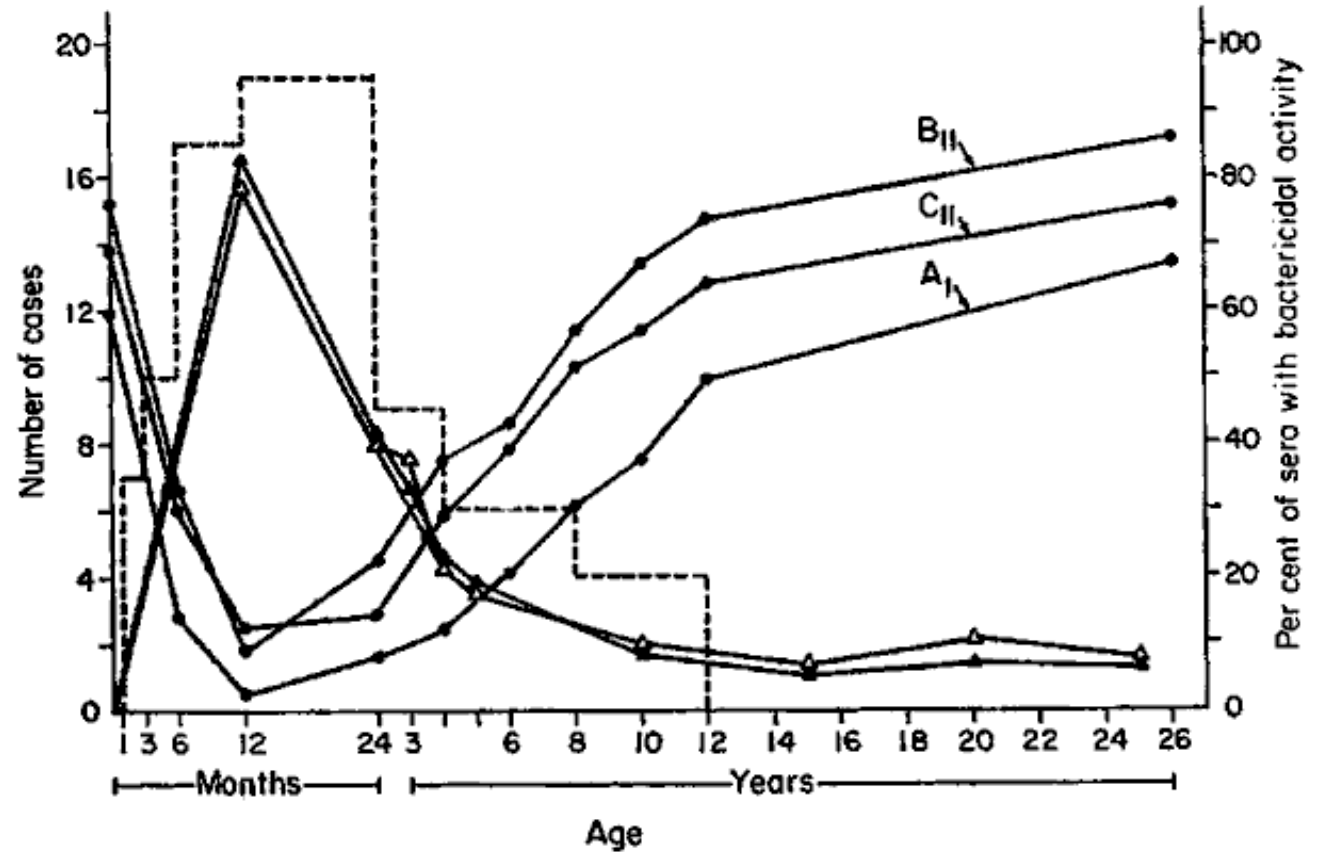


Brundage, J. F. & Zollinger, W. D. (1987). Evolution of meningococcal disease epidemiology in the US army. In Evolution of meningococcal disease, pp. 5-25. Edited by N. A. Vedros. Boca Raton, FL: CRC Press.

Wang, J. F., Caugant, D. A., Morelli, G., Koumaré, B. & Achtman, M. (1993). Antigenic and epidemiological properties of the ET-37 complex of *Neisseria meningitidis*. *J Infect Dis.* **167**, 1320-1329.

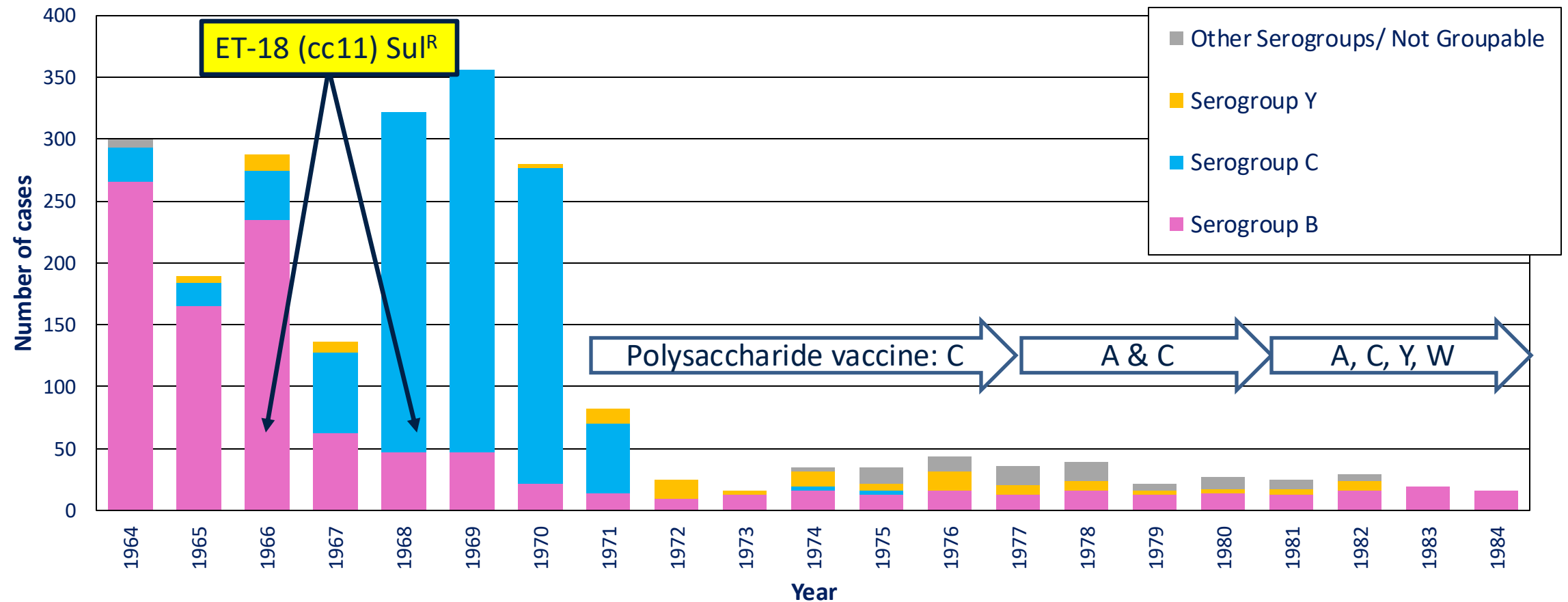
Meningococcal polysaccharide vaccines: establishing the paradigm

- Surveys of bactericidal activity of subjects' blood, compared with age-specific disease incidence.
- The inverse relationship, and loss of maternal immunity, suggests antibody-mediated immunity is important.
- This led to the development of the **plain polysaccharide** vaccines.



Goldschneider, I., Gotschlich, E. C. & Artenstein, M. S. (1969). Human Immunity to the Meningococcus. II. Development of Natural Immunity. *J Exp Med* **129**, 1327-1348.

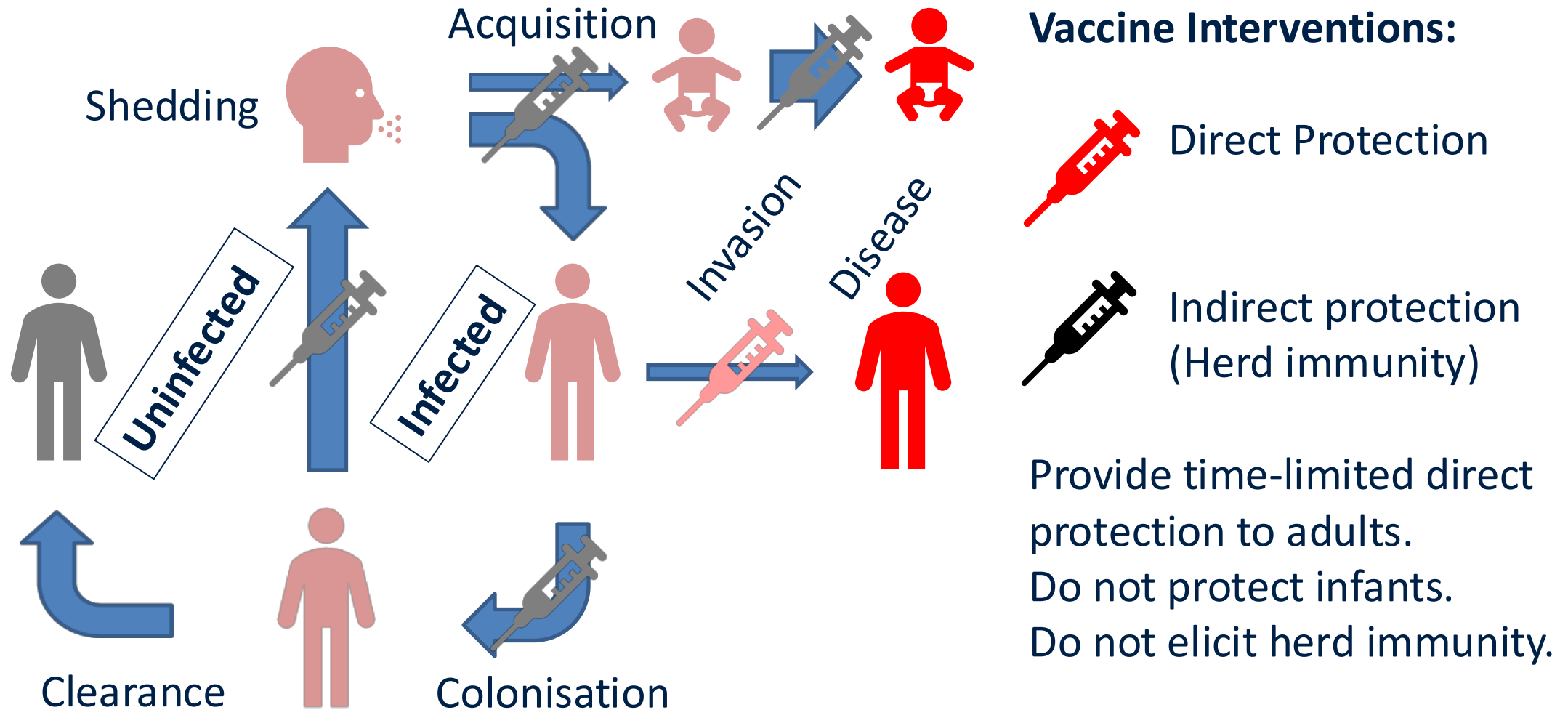
USA Army Outbreaks: Sufonamide resistance and the development of plain polysaccharide vaccines



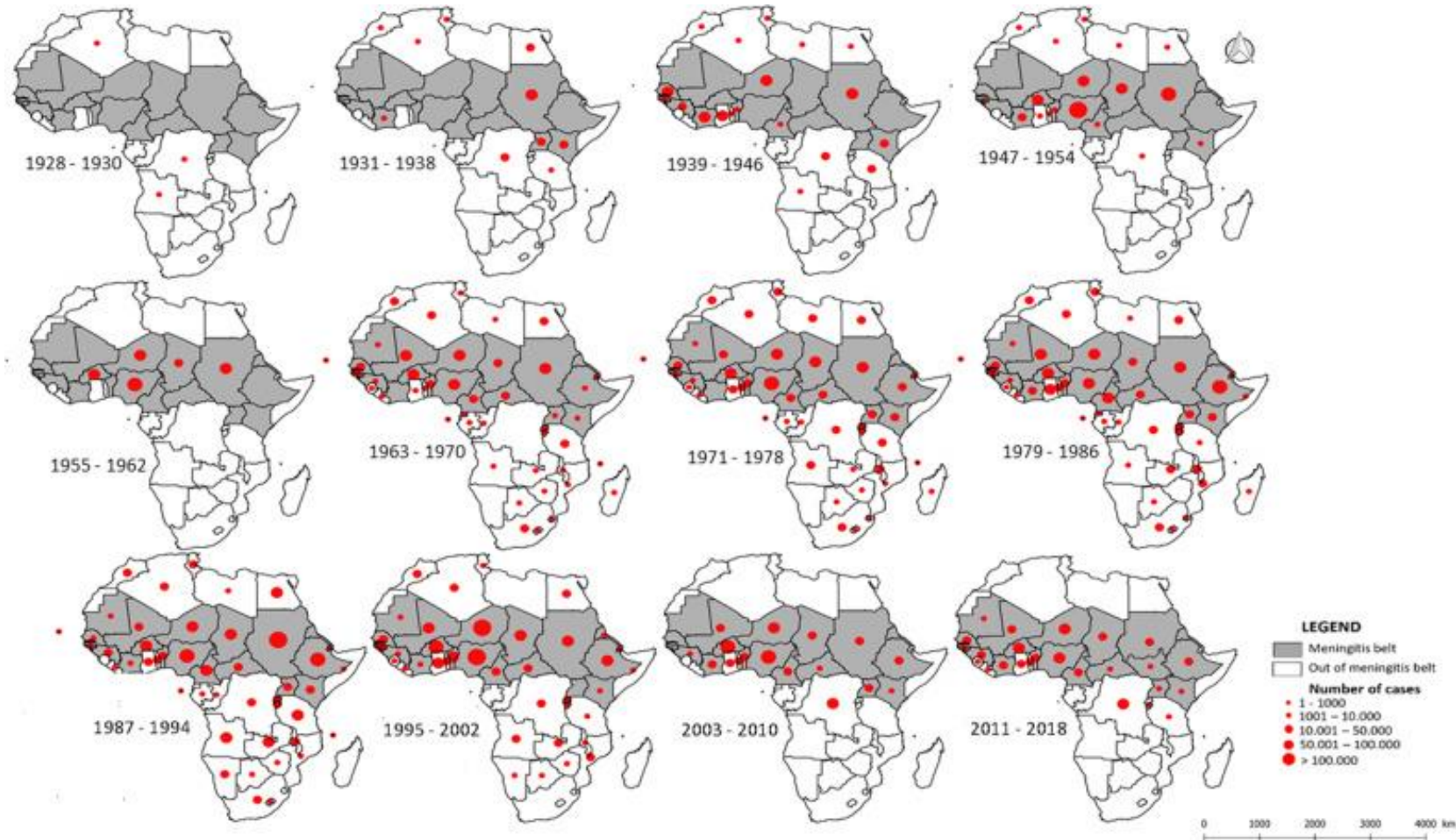
Brundage, J. F. & Zollinger, W. D. (1987). Evolution of meningococcal disease epidemiology in the US army. In Evolution of meningococcal disease, pp. 5-25. Edited by N. A. Vedros. Boca Raton, FL: CRC Press.

Wang, J. F., Caugant, D. A., Morelli, G., Koumaré, B. & Achtman, M. (1993). Antigenic and epidemiological properties of the ET-37 complex of *Neisseria meningitidis*. *J Infect Dis.* **167**, 1320-1329.

'Plain' polysaccharide vaccines (pre-1999)

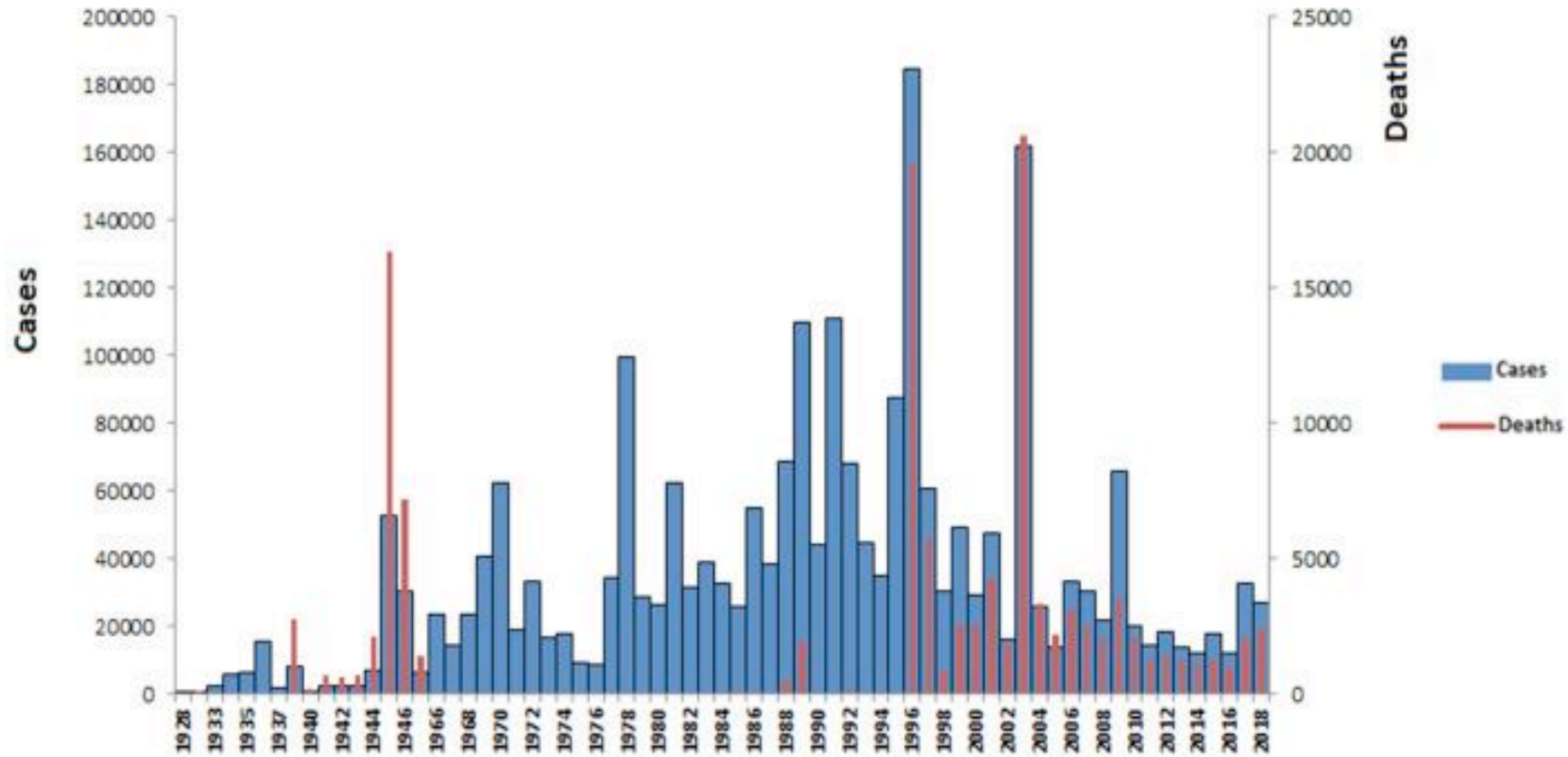


Meningococcal disease in Africa



Mazamay, S., Guegan, J. F., Diallo, N., Bompangue, D., Bokabo, E., Muyembe, J. J., Taty, N., Vita, T. P. & Broutin, H. (2021). An overview of bacterial meningitis epidemics in Africa from 1928 to 2018 with a focus on epidemics "outside-the-belt". *BMC Infect Dis.* **21**, 1027.

Meningococcal disease in Africa



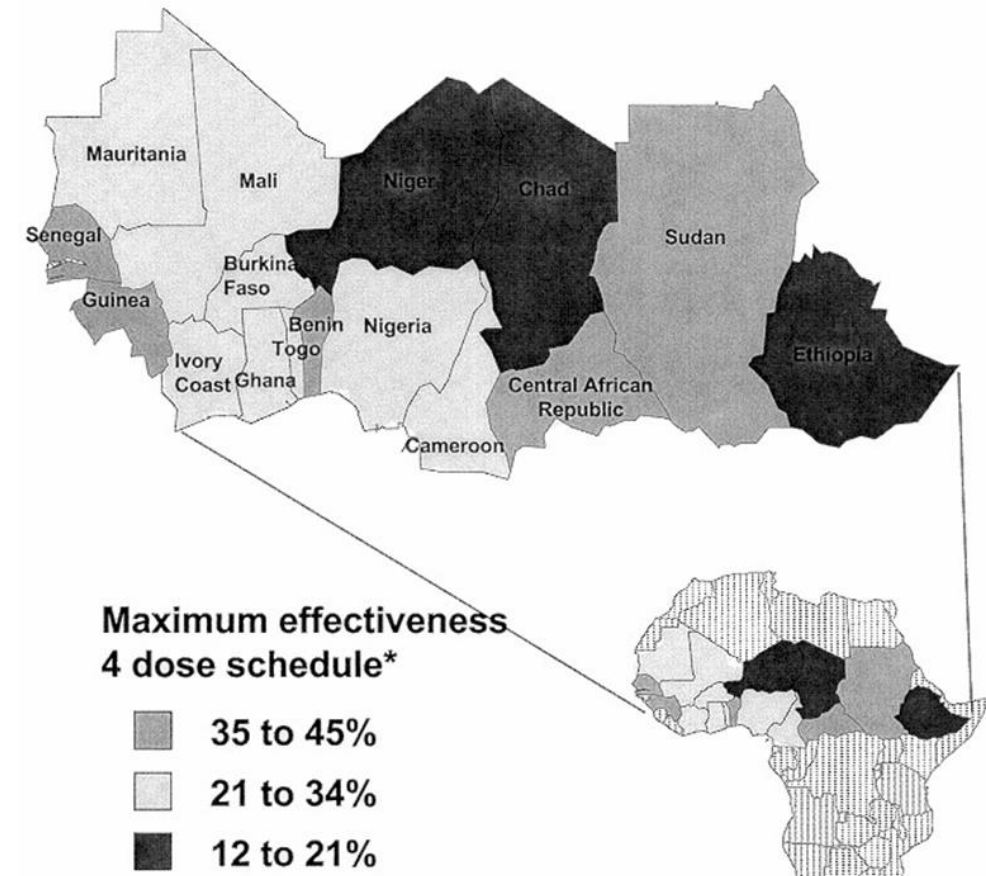
Mazamay, S., Guegan, J. F., Diallo, N., Bompangue, D., Bokabo, E., Muyembe, J. J., Taty, N., Vita, T. P. & Broutin, H. (2021). An overview of bacterial meningitis epidemics in Africa from 1928 to 2018 with a focus on epidemics "outside-the-belt". *BMC Infect Dis.* **21, 1027.**

'Plain' polysaccharide vaccines in Africa

Table 2. Number of serogroup A meningococcal meningitis cases in the vaccine and vaccine control groups

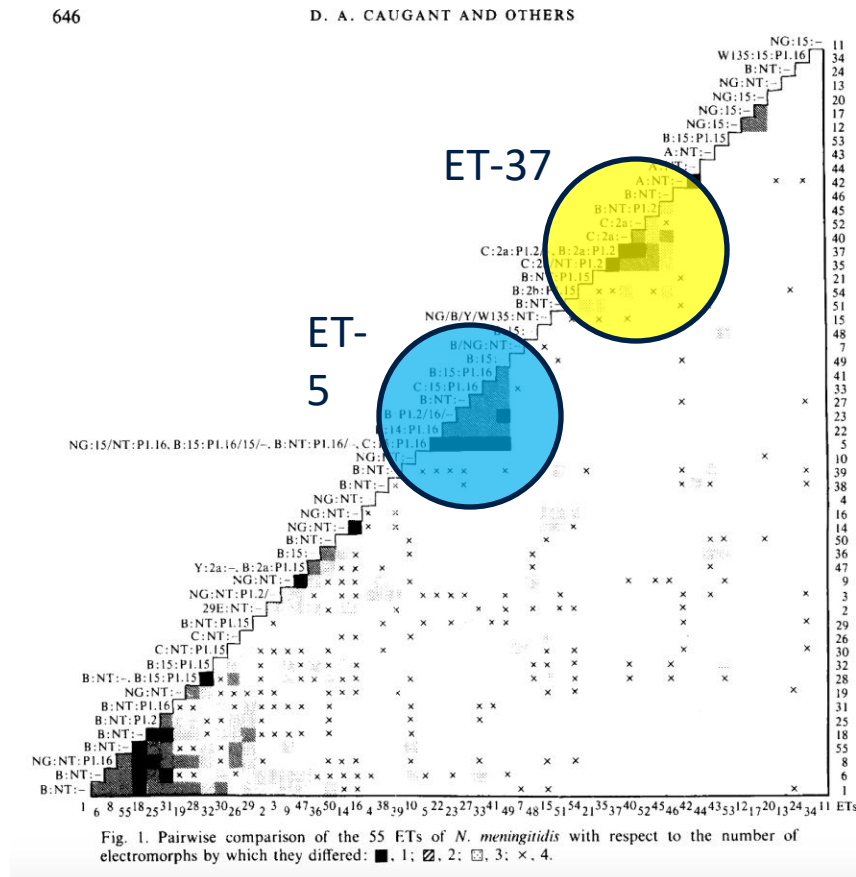
Area	Vaccine employed			
	PAV		TTC	
	No. vaccinated	CSM cases	No. vaccinated	CSM cases
Khartoum South	7 966	0	7 950	6
Mahdia	2 925	0	2 799	1
Total	10 891	0	10 749	7

Erwa, H. H., Haseeb, M. A., Idris, A. A., Lapeyssonnie, L., Sanborn, W. R., and Sippel, J. E. (1973) A serogroup A meningococcal polysaccharide vaccine: studies in the Sudan to combat cerebrospinal meningitis caused by *Neisseria meningitidis* group A. *Bull World Health Organ* **49**, 301-305



Miller, M. A., Wenger, J., Rosenstein, N., and Perkins, B. (1999) Evaluation of meningococcal meningitis vaccination strategies for the meningitis belt in Africa. *Pediatr Infect Dis J* **18**, 1051-1059

1986:Meningococcal phenotype to genotype: hyperinvasive variants

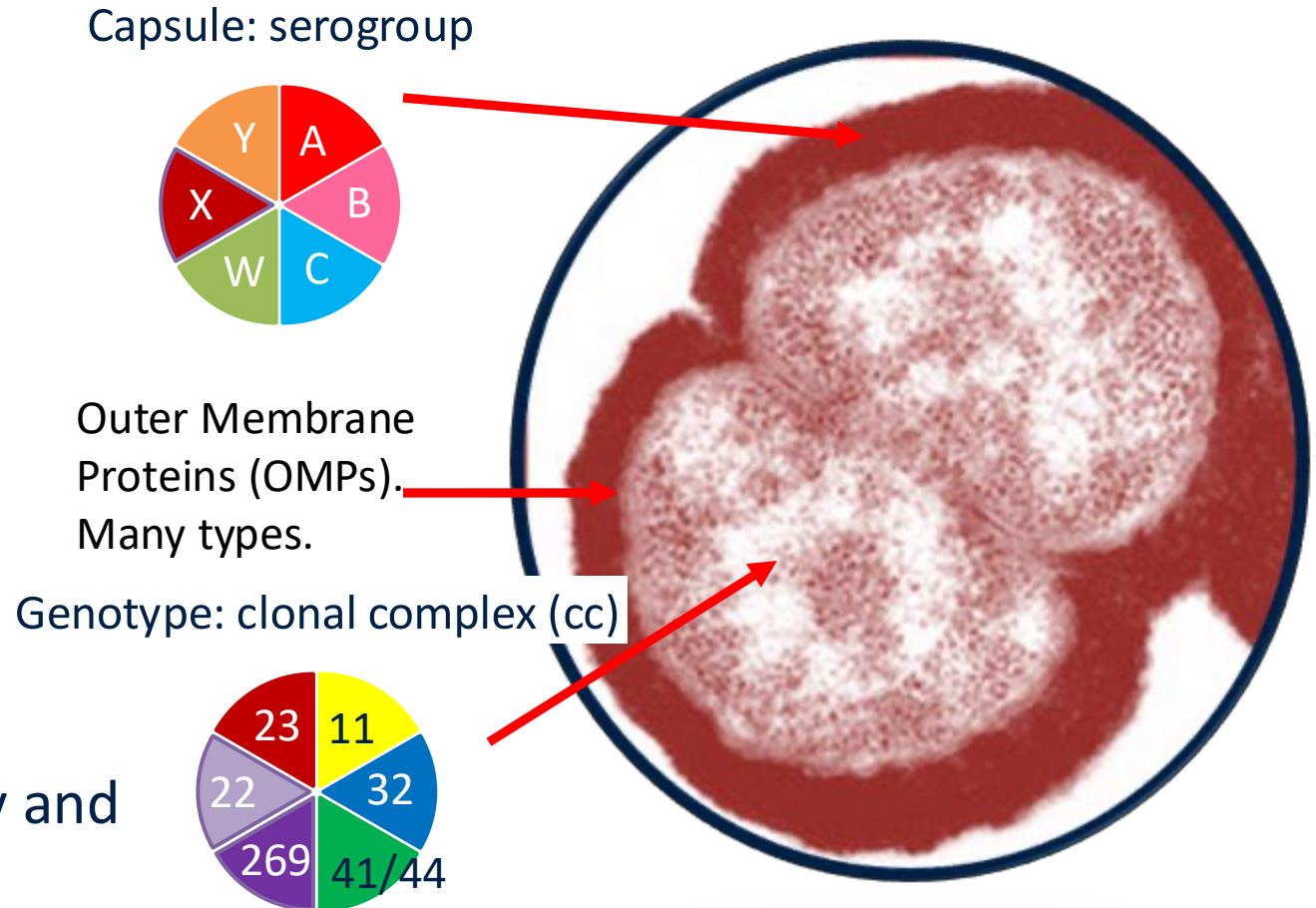


Caugant, D. A., Bovre, K., Gaustad, P., Bryn, K., Holten, E., Høiby, E. A. & Frøholm, L. O. (1986). Multilocus genotypes determined by enzyme electrophoresis of *Neisseria meningitidis* isolated from patients with systemic disease and from healthy carriers. *J Gen Microbiol.* **132, 641-652.**



Neisseria meningitidis, the meningococcus

- Gram negative bacterium,
 - two-layer cell wall.
- Diplococcus,
 - takes the form of two spheres.
- Can be covered by a capsule,
 - A sugar coat that protects it.
- Very diverse,
 - many different types genetically and antigenically.



Ganesh, K., Allam, M., Wolter, N., Bratcher, H. B., Harrison, O. B., Lucidarme, J., Borrow, R., de Gouveia, L., Meiring, S., Birkhead, M., Maiden, M. C., von Gottberg, A. & du Plessis, M. (2017). Molecular characterization of invasive capsule null *Neisseria meningitidis* in South Africa. *BMC Microbiology* **17**, 40.

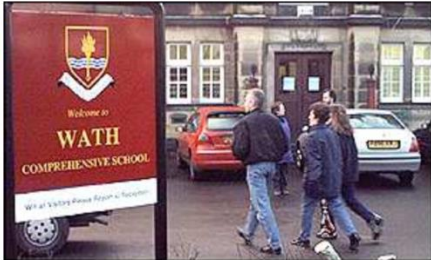
Conjugate vaccines: a new era for meningococcal disease prevention

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BBC NEWS

You are in: **Health**
Tuesday, 9 March, 1999, 18:01 GMT

Meningitis: The 1999 panic



William Hague's old school saw the first outbreak of the year

The government's vaccination programme was prompted by a series of serious meningitis outbreaks.

Throughout January 1999, a severe outbreak of meningitis in south Wales took a prominent position in the headlines.

As a public health emergency was declared, mass vaccinations were performed at schools in Pontypridd.

Doctors did their best to calm public fears about the likelihood of infection, but parents were alarmed and marched to demand blanket vaccinations - regardless of whether their children attended the affected schools.

See also:

- 09 Mar 99 | Medical notes
Meningitis: The argument for mass vaccinations
- 09 Mar 99 | Medical notes
Preventing meningitis
- 05 Jan 99 | Health
Killer disease claims more victims
- 18 Jan 99 | Health
Meningitis trauma service launched
- 05 Jan 99 | Health
Meningitis research targets teenagers
- 11 Feb 99 | Health
Meningitis study group set up

Internet links:

- PHLS facts and figures on meningococcal disease
- Meningitis Research Foundation
- National Meningitis Trust

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Top Health stories now:

- Postcode lottery in GP services
- IVF mix-up heads for court
- Transplant first for cancer patient
- Costly wait with dementia symptoms

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Monday, November 1, 1999 Published at 06:33 GMT

Health

Mass meningitis vaccination programme begins

The vaccine could prevent many deaths from the brain disease

The government has officially launched its meningitis vaccination programme, first targeting 15 to 17-year-olds and then babies.

Children around the UK are to become the first in the world to take part in routine vaccinations against the C strain of meningitis.

SPECIAL REPORT

THE MENINGITIS FILE

The vaccine has already been tested on more than 4,000 British children and 21,000 children outside the UK.

The mass immunisation programme has also been piloted in Ironville, South Derbyshire, which has been particularly affected by the brain disease.

Health Contents

- Background Briefings
- Medical notes
- Relevant Stories**
- 01 Nov 99 | Northern Ireland
Doctors frustrated over vaccine scarcity
- 20 Jul 99 | Health
Mass vaccine targets meningitis
- 12 Feb 99 | Medical notes
The argument for mass meningitis vaccinations
- 10 Feb 99 | Medical notes
Meningitis: Preventive measures

Internet Links

- National Meningitis Trust
- Meningitis Research Foundation
- Department of Health

The BBC is not responsible for the content of external internet sites.

In this section

- Disability in depth
- Spotlight: Bristol inquiry
- Antibiotics: A fading wonder

Miller, E., Salisbury, D. & Ramsay, M. (2001). Planning, registration, and implementation of an immunisation campaign against meningococcal serogroup C disease in the UK: a success story. *Vaccine* 20, S58-67.

1999: Meningococcal conjugate vaccines: new opportunities and new challenges

- Meningococcal C conjugate (MCC) vaccines to be introduced without phasing

- Most serogroups (ST-11, cc11)

- Meningococcal

- Important to MCC vaccine

transmission, and population biology.

“Carriage should be done among the 15-17 age group – i.e. those with high rates of carriage and who are to be included in the first round of vaccination.”



Brian Spratt

Maiden, M. C. J. & Spratt, B. G. (1999). Meningococcal conjugate vaccines: new opportunities and new challenges. *Lancet*. **354**, 615-616.

Carriage studies



UK Meningococcal Carriage Study

Today's Date / / Name of School

**** This information is completely confidential. It will not be seen by school staff or other students. ****

Q1 What is your current age in years? 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐

Q2 I am: male ☐ female ☐

Q3 What is your home postcode?

Q4 Which school year are you in? S5 ☐ S6 ☐

Q5 Do you currently have a cold or sore throat? NO ☐ YES ☐

Q6 Are you currently taking or have you recently stopped taking antibiotics?
not taken in the past month ☐ stopped in the last month ☐ stopped in the last week ☐ yes, currently taking ☐

Q7 How many cigarettes do you smoke in a typical day?
0 ☐ 1-5 ☐ 6-10 ☐ 11-20 ☐ more than 20 ☐

Q8 How many times have you smoked an e-cigarette in the last week?
0 ☐ 1-2 ☐ 3-4 ☐ 7 or more ☐

Q9 How many times have you smoked a waterpipe (shisha, hookah, hubbly bubbly) in the last month?
0 ☐ 1 ☐ 2 ☐ 3-4 ☐ 5 or more ☐

Q10 Does any other person at home smoke cigarettes? NO ☐ YES, outside the house ☐
YES, inside the house ☐

Q11 How many days in the last week have you been to a party, pub, bar or night club?
0 ☐ 1 ☐ 2-3 ☐ 4-5 ☐ 6 ☐

Q12 How many people have you kissed (kissing with tongues, not just lips or cheeks) in the last week?
0 ☐ 1 ☐ 2-3 ☐ 4 or more ☐

Q13a Do you have a regular girlfriend or boyfriend? NO ☐ YES ☐

Q13b If YES: do they smoke cigarettes? NO ☐ YES ☐
do they smoke shisha? NO ☐ YES ☐

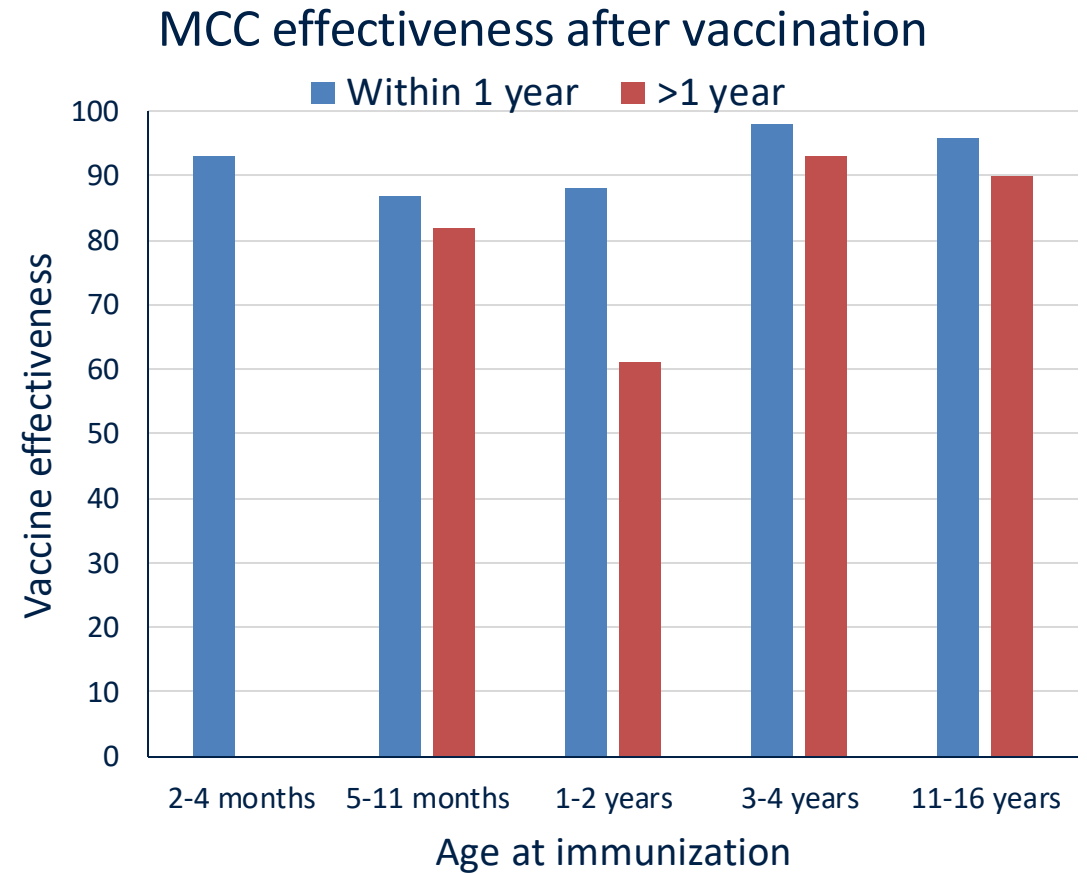
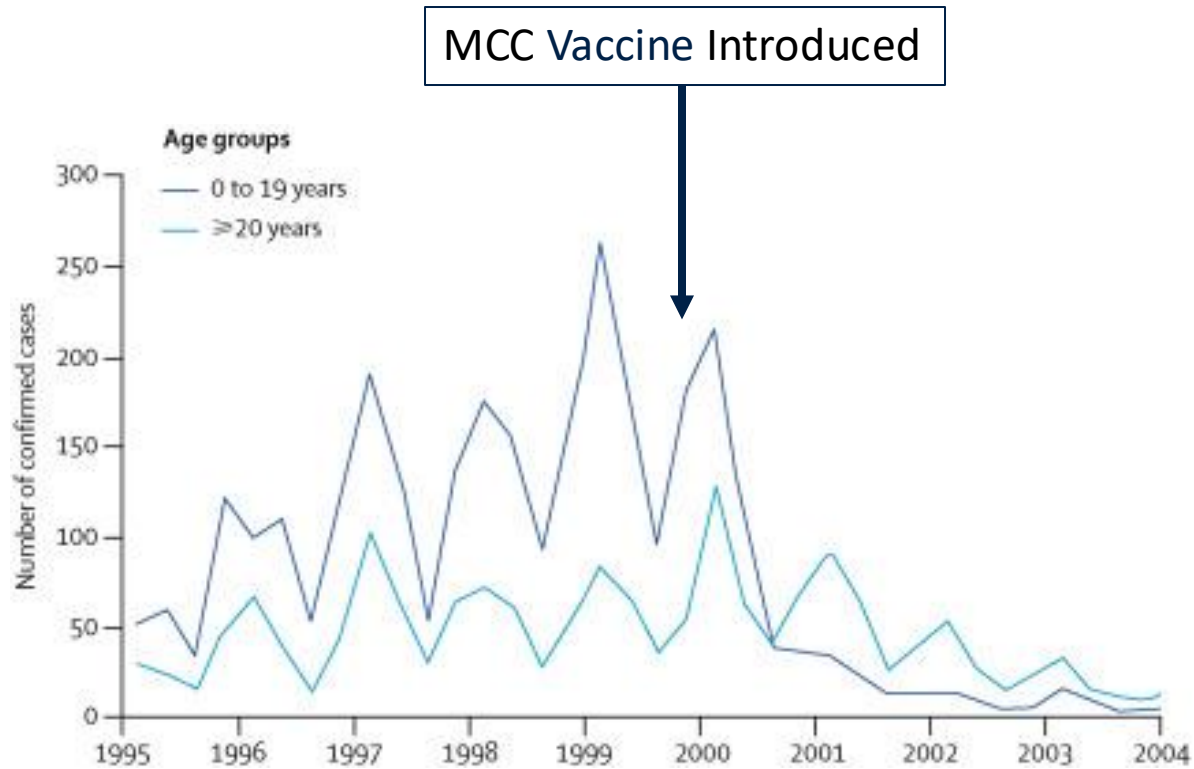
Q14 What is your ethnic group?
White ☐ Asian/Asian British ☐ Black/African/Caribbean/Black British ☐
Mixed/multiple ethnic ☐ Other ethnic group ☐

Thank you for completing this questionnaire

UKMENCAR4 Questionnaire v1.2 11/08/14 REC REF 14/SC/1163

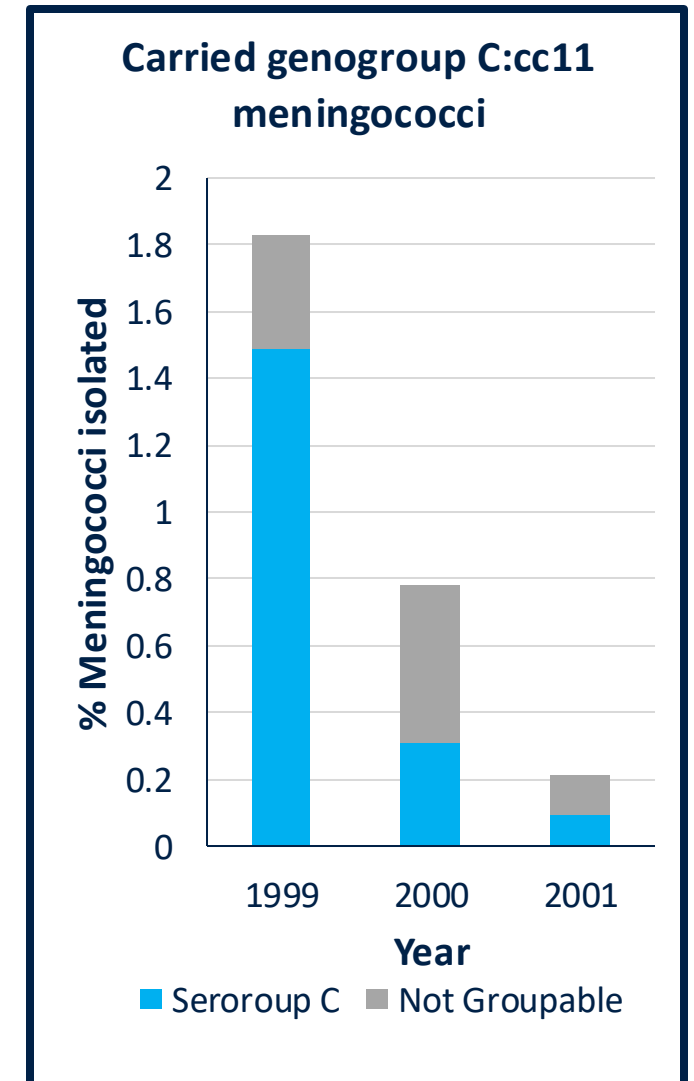
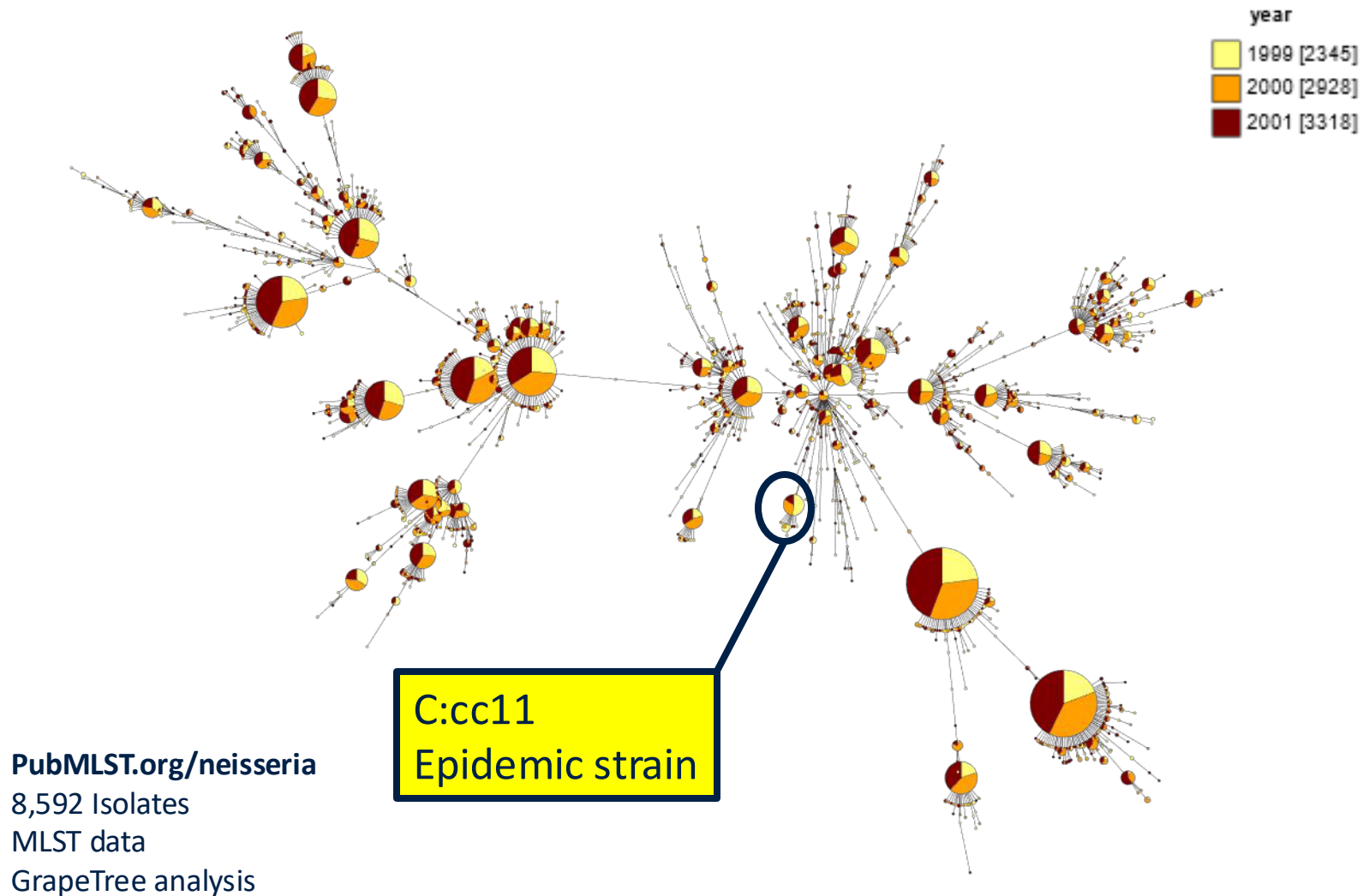
Bratcher, H. B., Rodrigues, C. M. C., Finn, A., Wootton, M., Cameron, J. C., Smith, A., Heath, P., Ladhani, S., Snape, M. D., Pollard, A. J., Cunningham, R., Borrow, R., Trotter, C., Gray, S. J., Maiden, M. C. J. & MacLennan, J. M. (2019). UKMenCar4: A cross-sectional survey of asymptomatic meningococcal carriage amongst UK adolescents at a period of low invasive meningococcal disease incidence. *Wellcome Open Res* 4, 118.

MCC vaccination & serogroup C disease: England and Wales



Trotter, C. L., N. J. Andrews, E. B. Kaczmarski, E. Miller, and M. E. Ramsay. (2004). Effectiveness of meningococcal serogroup C conjugate vaccine 4 years after introduction. *Lancet* **364**, 365-7.

UK meningococcal carriage studies, 1999-2001

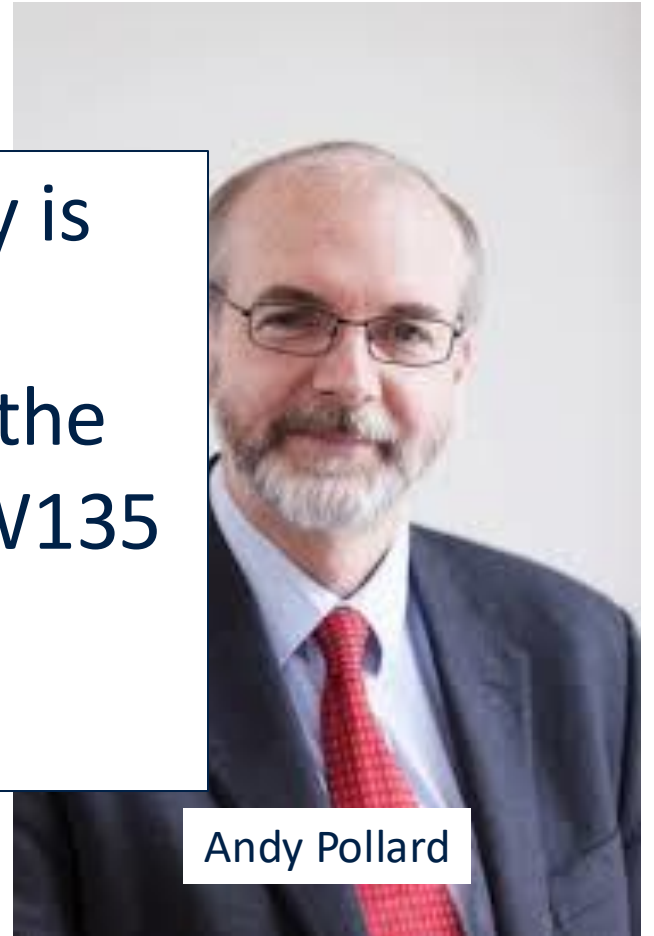


2003: Epidemic meningococcal disease in sub-Saharan Africa—towards a sustainable solution?

- History of epidemic disease in Africa since 1905
 - calls for W(-135) vaccination at time of writing.

- Reactive vaccination
 - Problems with
- Meningococcal disease
 - serogroups
 - Dynamics, remains poorly understood
- A multivalent vaccine needed,
 - currently monovalent proposed.

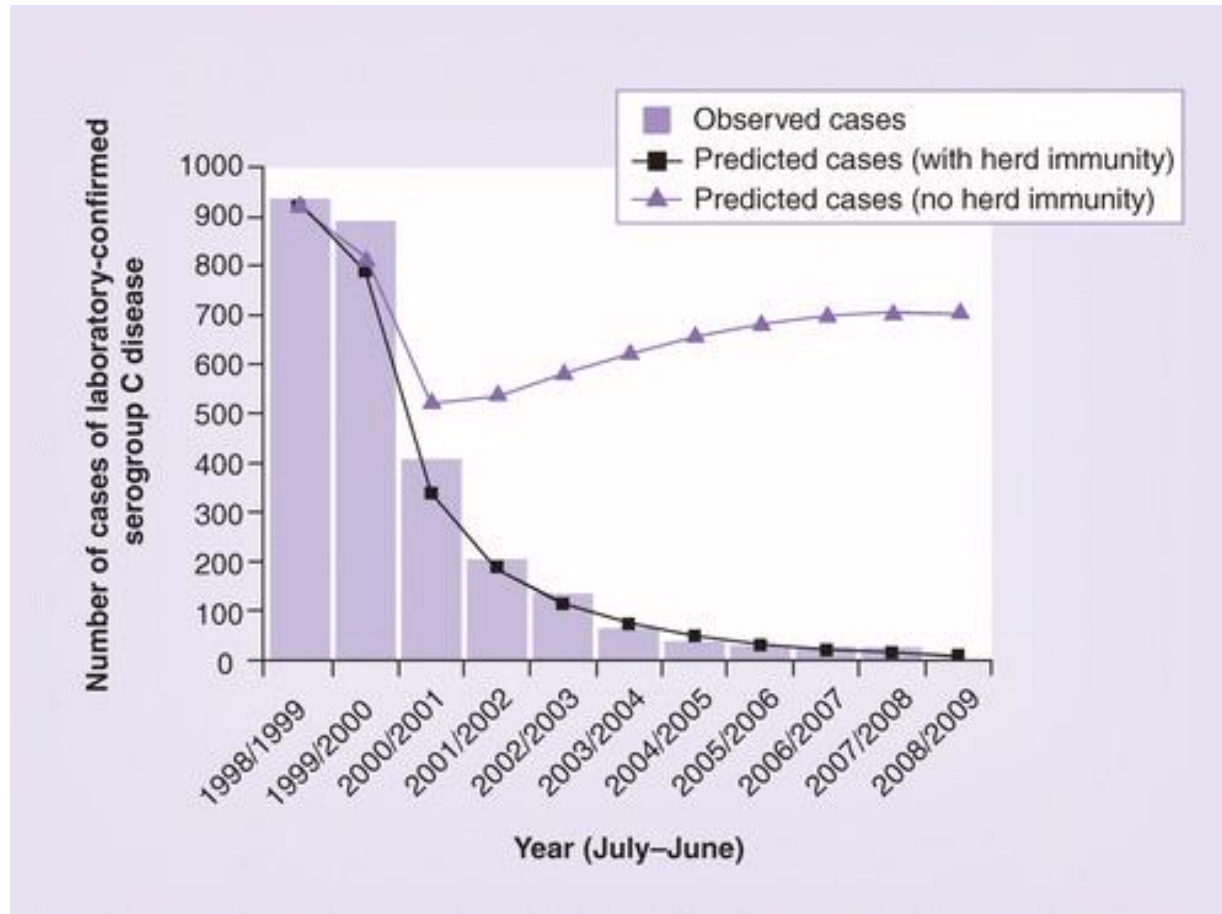
“The tragedy of this unfolding story is that the technology necessary for the production of a vaccine for the sustainable control of A, C, Y, and W135 meningococcal disease has been available for more than a decade.”



Andy Pollard

Pollard, A. J. & Maiden, M. C. (2003). Epidemic meningococcal disease in sub-Saharan Africa--towards a sustainable solution? *Lancet Infectious Diseases*. **3**, 68-70.

2009: Meningococcal vaccines and herd immunity



Trotter, C. L. & Maiden, M. C. (2009). Meningococcal vaccines and herd immunity: lessons learned from serogroup C conjugate vaccination programs. *Expert Review Vaccines*. **8**, 851-861.

2010: MenAfriVac® a solution to epidemic meningococcal disease in Africa

Meningitis Vaccine Project (MVP):

- Gates funded, partnership with WHO and PATH;
- Innovative development – Northern technology, southern manufacturer;
- Low cost (less than \$1 per dose, for sustainable use);
- MenAfriVac, a polysaccharide-protein conjugate vaccine, introduced 2010 in Burkina Faso;
- **Everyone aged 1-29 years immunised;**
- **Immediate reduction in disease levels.**

Diomande, F. V. K., Djingarey, M. H., Daugla, D. M., Novak, R. T., Kristiansen, P. A., Collard, J. M., Gamougam, K., Kandolo, D., Mbakuliyemo, N., Mayer, L., Stuart, J., Clark, T., Tevi-Benissan, C., Perea, W. A., Preziosi, M. P., LaForce, F. M., Caugant, D., Messonnier, N., Walker, O. & Greenwood, B. (2015). Public Health Impact After the Introduction of PsA-TT: The First 4 Years. *Clinical Infectious Diseases* 61, S467-S472.



Samba Sow &
first MenAfriVaccinee



Marc La Force

Chad outbreak, 2011-2012

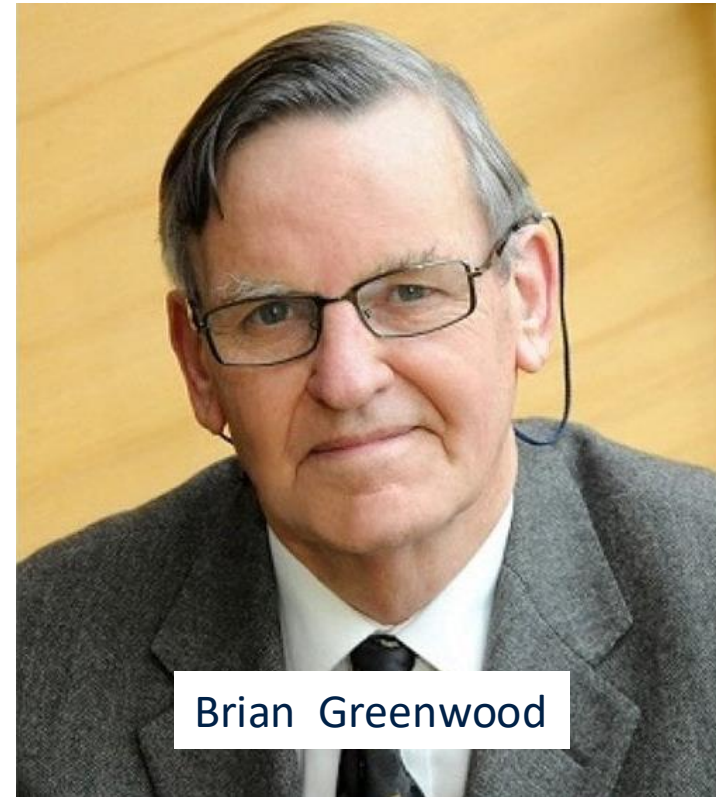
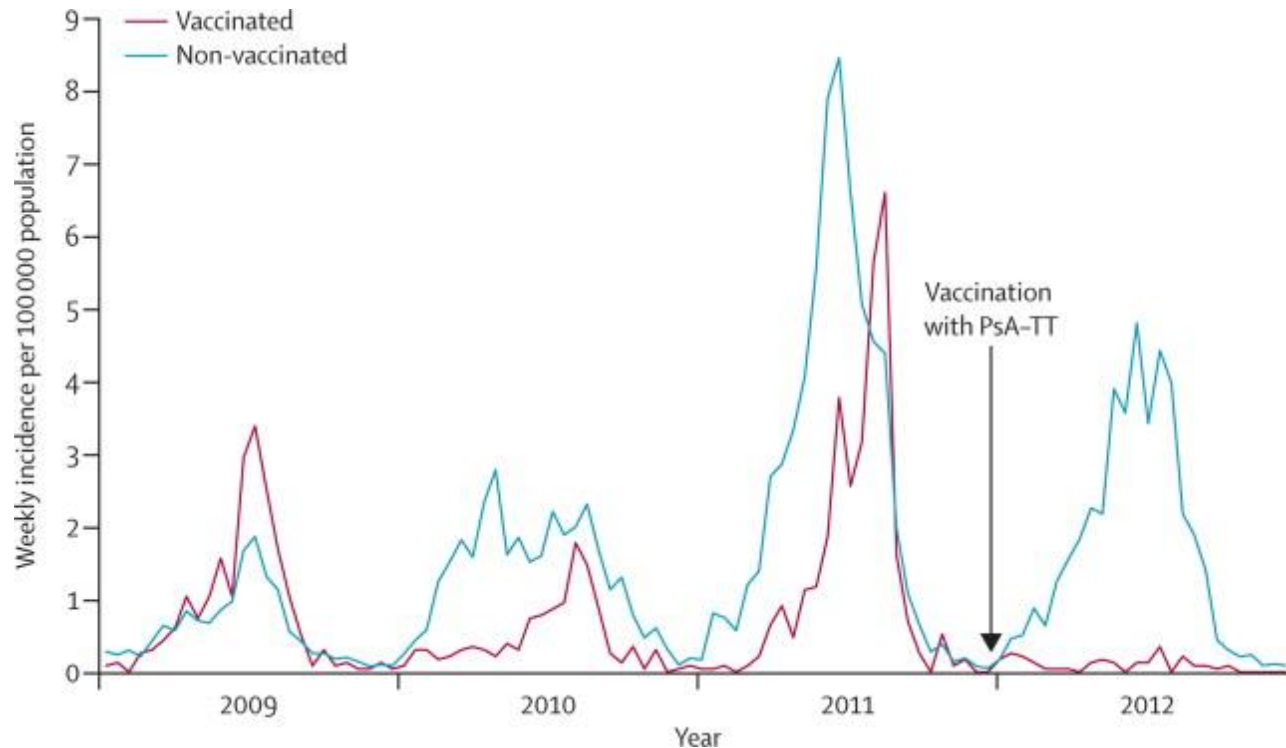
MenAfriVac™ vaccine in the village of Sari, Chad, December 2011



2011-12: MenAfriVac® in Chad



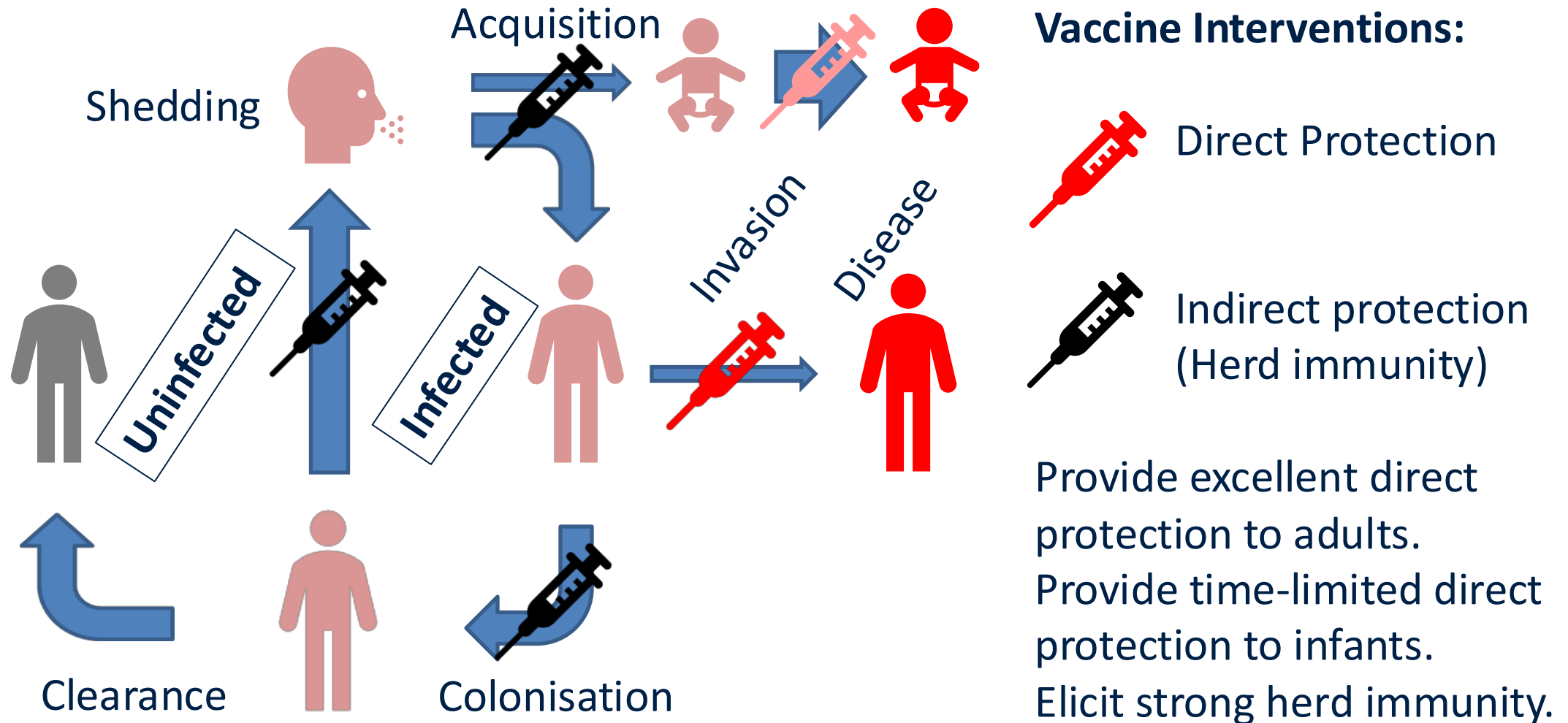
a) Disease in Chad



Brian Greenwood

Daugla, D. M., Gami, J. P., Gamougam, K., Naibei, N., Mbainadji, L., Narbe, M., Toralta, J., Kodbesse, B., Ngadoua, C., Coldiron, M. E., Fermon, F., Page, A. L., Djingarey, M. H., Hugonnet, S., Harrison, O. B., Rebbetts, L. S., Tekletsion, Y., Watkins, E. R., Hill, D., Caugant, D. A., Chandramohan, D., Hassan-King, M., Manigart, O., Nascimento, M., Woukeu, A., Trotter, C., Stuart, J. M., Maiden, M. C. & Greenwood, B. M. (2014). Effect of a serogroup A meningococcal conjugate vaccine (PsA-TT) on serogroup A meningococcal meningitis and carriage in Chad: a community study. *Lancet* **383**, 40-47.

Protein conjugate polysaccharide vaccines



2012: Can we, should we, eliminate the meningococcus?

- Almost certainly neither achievable nor desirable,
 - control, elimination, or eradication of invasive meningococci more likely.

- MenaAfriVac disease,
 - other serog
 - Serogroups
 - serogroup

- Most problem meningococci
 - potential v
 - complexes.

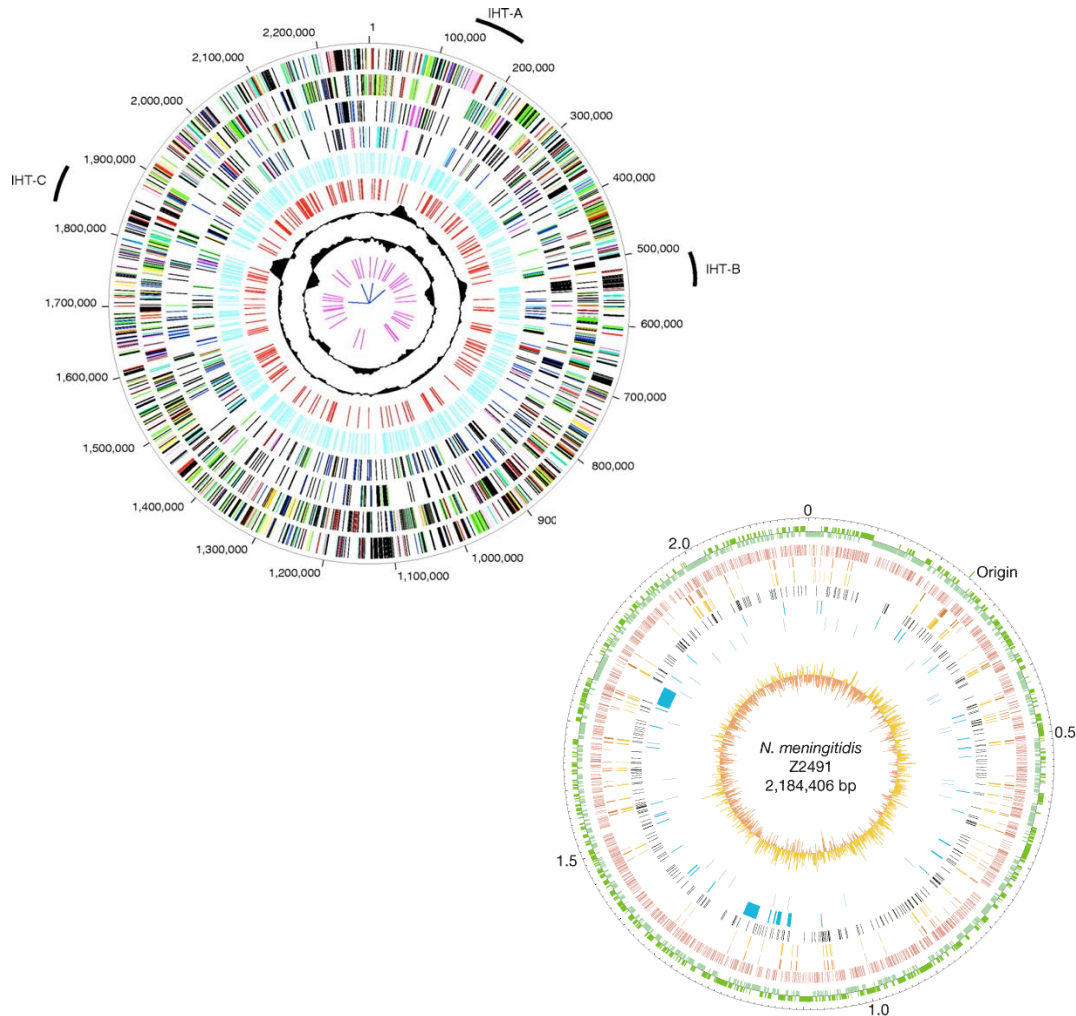
- A world free of invasive meningococci a distant prospect
 - although less meningococcal disease achievable and worthwhile.

“The eradication of all carried meningococci is almost certainly neither achievable nor desirable: the control, elimination, or eradication of particular invasive meningococci is a more realistic goal.”

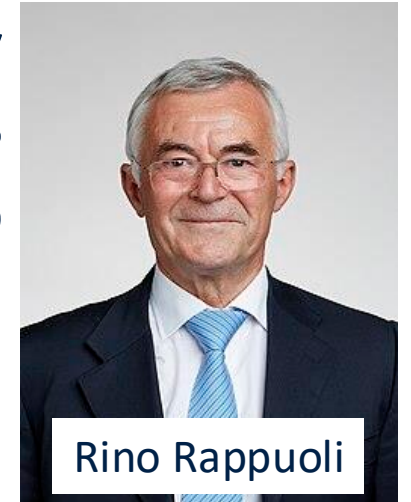


Matthias Frosch

Novel approaches to group B vaccines: a tale of 2 genomes



Reverse Vaccinology
BexSero, 4CMenB
FHbp



Conventional Vaccinology
Trumenba
Lp2086

2011: Meningococcal serogroup B vaccines: will they live up to expectations?

Outer Membrane Vesicle vaccines:

- Are based on the outer membrane 'blebs' produced by meningococci in culture;
- contain many antigens including LOS and proteins.
- the PorA protein is a major component.
- are variant specific.
- are not known to affect carriage.

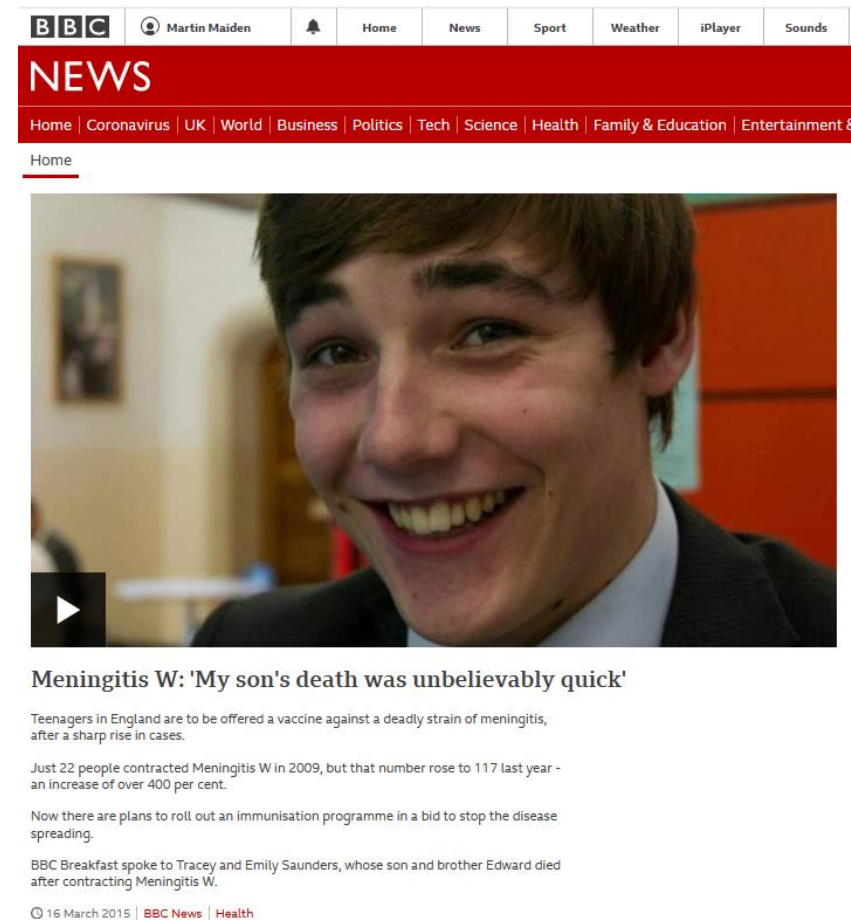


Zollinger, W. D., Poolman, J. T. & Maiden, M. C. J. (2011). Meningococcal serogroup B vaccines: will they live up to expectations? *Expert Review of Vaccines*. **10**, 559-561.

2009: The return of cc11, W:cc11 in the UK

- After MCC introduction, overall reductions in IMD.
- W:cc11 IMD began to rise in 2009.
- Rapid rise associated with atypical presentations, including:
 - Pneumonia (12%);
 - Septic arthritis (7%);
 - Epiglottitis/supraglottitis (4%).
- High rates of ICU admission (37%) & fatality (12%).

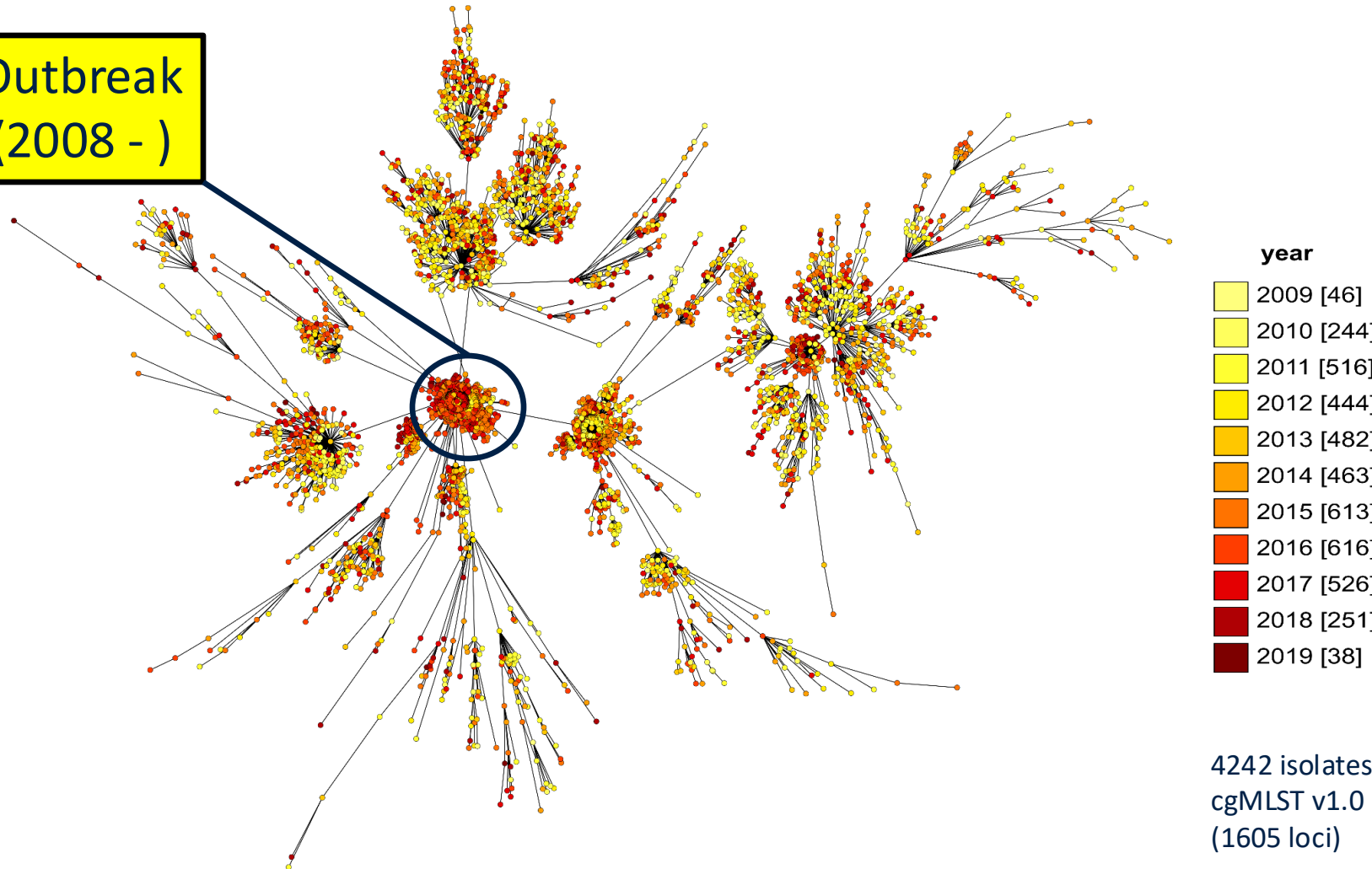
Ladhani, S. N., Beebeejaun, K., Lucidarme, J., Campbell, H., Gray, S., Kaczmarek, E., Ramsay, M. E. & Borrow, R. (2015). Increase in Endemic *Neisseria meningitidis* Capsular Group W Sequence Type 11 Complex Associated With Severe Invasive Disease in England and Wales. *Clinical Infectious Diseases*. **60**, 578-585.



<https://www.bbc.co.uk/news/av/health-31904057>

2009-2019: Disease-causing meningococci in the UK

W:c11 Outbreak
variant (2008 -)



Vaccine policy change 2015

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Health

Meningitis vaccine plan after steep rise in new strain

By James Gallagher
Health editor, BBC News website

13 March 2015 | Health



Teenagers will soon be vaccinated against deadly meningitis W after a steep rise in the number of cases, Public Health England has announced.

There were 22 confirmed cases in 2009, 117 last year and experts predict even more cases in the future.

Meningitis W also has a higher-than-usual death rate.

The government's Joint Committee on Vaccination and Immunisation called for 14 to 18-year-olds to be vaccinated "as soon as possible".

<https://www.bbc.co.uk/news/health-31869055>

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Features

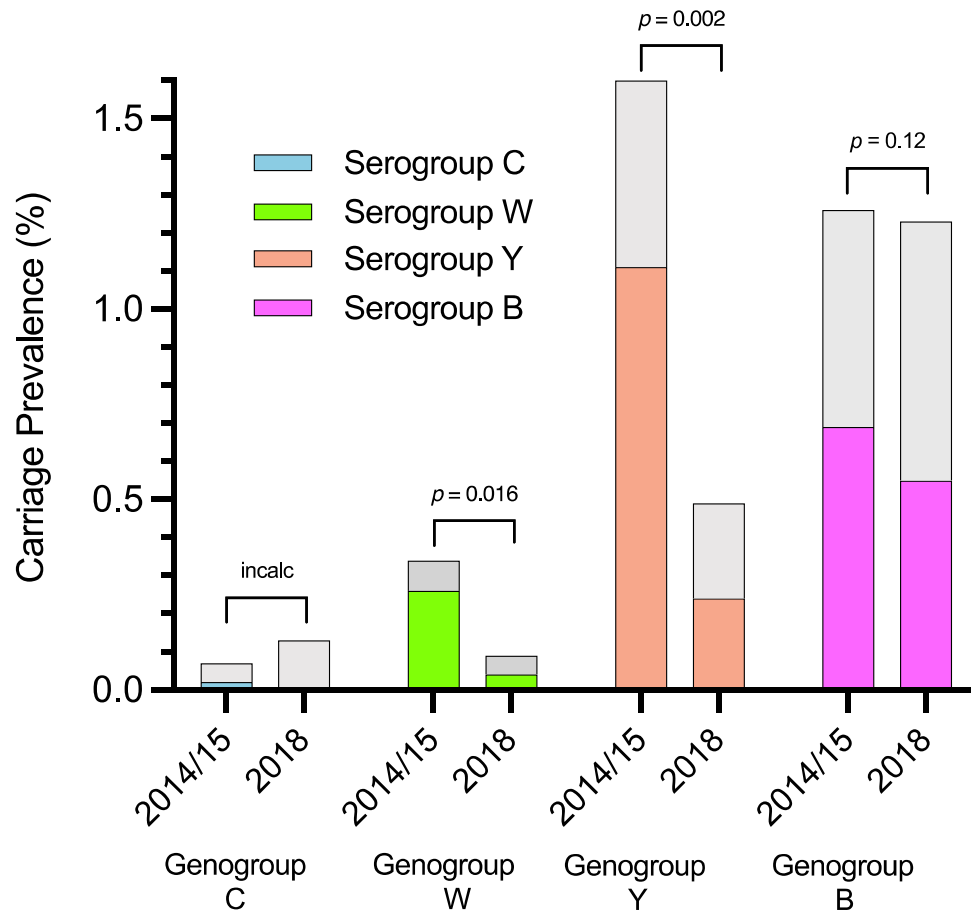
Fatal silence
Why do so many fortysomething men kill themselves?



- In 2015 the UK JCVI recommended the replacement of the teenage MCC booster with MenACWY conjugate vaccine.
- Aimed to combat the spread of the 'South American' variant by a combination of direct and indirect (herd) protection.

Ladhani, S. N., Ramsay, M., Borrow, R., Riordan, A., Watson, J. M. & Pollard, A. J. (2016). Enter B and W: two new meningococcal vaccine programmes launched. *Arch Dis Child* 101, 91-95.

Meningococcal carriage in UK teenagers



- Carried meningococci before (2014/15) and after (2018) Men ACWY conjugate vaccine introduction (09/2015):
 - Men B, unchanged;
 - Men C, remained very low;
 - Men W and Men Y, significantly reduced.

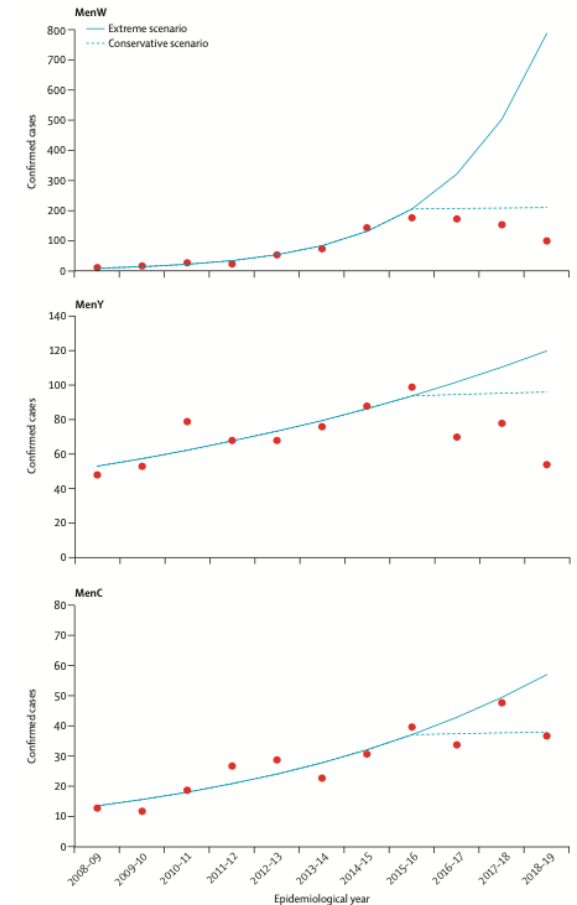
Carr, J., MacLennan, J. M., et al., C., Maiden, A., Snape, M. D. (2022). Impact of meningococcal ACWY conjugate vaccines on pharyngeal carriage in adolescents: evidence for herd protection from the UK MenACWY programme . *Clin Microbiol Infect.* 28, P1649.e1641-1649.e1648.

Social behavior associated with meningococcal carriage over time

Risk factor	1999 % self-reporting	UKMenCar4 2014/15 % self-reporting	TEAM 2018 W1 & 2 % self-reporting
Cigarette smoker	22.9	11.3	6.8
6-10 cigarettes/day	7.3	3.0	5.1
11-20 cigarettes/day	3.8	1.5	1.2
>20 cigarettes/day	0.3	0.3	0.1
Intimate kissing last week	44.7	37.4	28.6
2-3 partners	4.8	2.9	2.0
4 or more partners	2.3	1.7	0.7
Pub or nightclub attendance last week	62.9	39.0	29.5
2-3 times	25.3	10.8	7.8
>3 times	8.3	1.7	1.1
Household smoking	37.5	27.4	20.2

Impact of UK MenACWY introduction

- MenW and MenY cases plateaued and then declined, MenC cases remained low.
- Significant reductions in cases in 14–18 year olds:
 - MenW (incidence rate ratio [IRR] 0.35 [95% CI 0.17–0.76])
 - MenY (0.21 [0.07–0.59]);
 - a non-significant reduction in MenC (0.11 [0.01–1.01]);
 - 205–1193 MenW cases prevented by indirect protection, 25 by direct protection.
 - 60–106 MenY, cases prevented by indirect protection and 19 by direct protection.
- Overall vaccine effectiveness against MenCWY disease was 94% (95% CI 80–99).



Campbell, H., Andrews, N., Parikh, S. R., White, J., Edelstein, M., Bai, X., Lucidarme, J., Borrow, R., Ramsay, M. E. & Ladhani, S. N. (2022). Impact of an adolescent meningococcal ACWY immunisation programme to control a national outbreak of group W meningococcal disease in England: a national surveillance and modelling study. *Lancet Child Adolesc Health*. 6, 96-105.

2017: Recent Progress in the Prevention of Serogroup B Meningococcal Disease

- Conjugate polysaccharide vaccines.
 - Effective against disease and carriage.
 - Multiple vaccines available;
 - Four serogroups covered (A,C,W,Y) and a fifth (X) in prospect.
- Protein based vaccines:
 - OMV, OMV/FHbp, FHbp;
 - Effective against some disease-causing meningococci, ineffective against carriage



Feavers, I. M. & Maiden, M. C. (2017). Recent progress in the prevention of serogroup B meningococcal disease. *Clinical and vaccine immunology : CVI*. **24**, e00566-00516.

2021: Impact of a Meningococcal Protein-based Serogroup B Vaccine on Serogroup W Invasive Disease in Children

Table 2. Analysis of Primary and Secondary Outcomes for *N. meningitidis* Carriage and Acquisition at 12 Months with the Use of Multiple Imputation.*

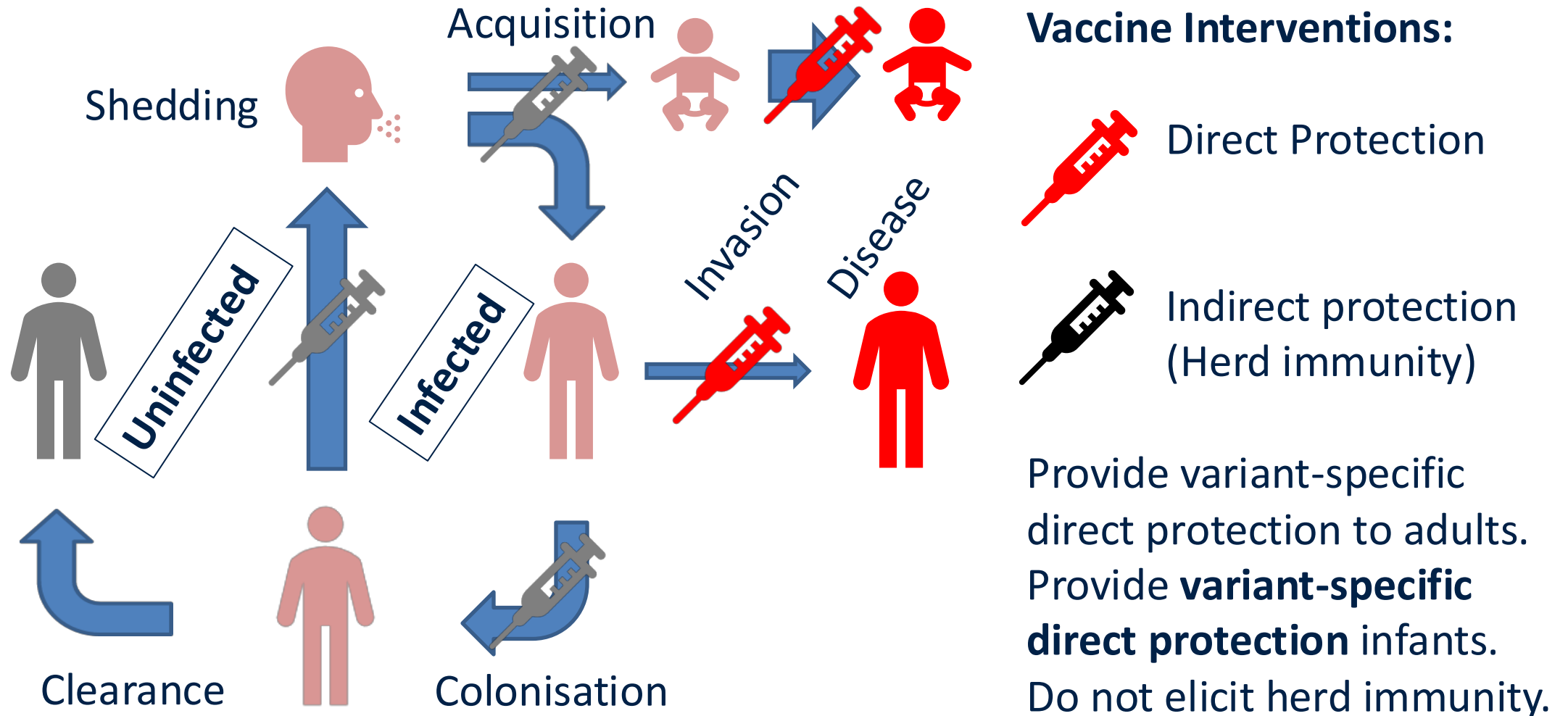
Outcome	Vaccination Group (N=12,746)	Control Group (N=11,523)	Odds Ratio (95% CI) [†]
	no. (%)		
Carriage of disease-causing genogroup	326 (2.55)	291 (2.52)	1.02 (0.80–1.31) [‡]
Carriage of any <i>N. meningitidis</i>	547 (4.29)	561 (4.87)	0.85 (0.70–1.04)
Carriage of genogroup B	164 (1.29)	135 (1.18)	1.10 (0.81–1.47)
Carriage of genogroup Y	117 (0.92)	131 (1.13)	0.81 (0.56–1.18)
Carriage of genogroup W§	17 (0.16)	18 (0.18)	0.89 (0.43–1.85)
Carriage of genogroup C§	12 (0.11)	7 (0.07)	1.87 (0.63–5.55)
Carriage of genogroup X§	8 (0.07)	1 (0.01)	7.59 (0.98–58.83) [¶]
Acquisition of any <i>N. meningitidis</i>	430 (3.38)	427 (3.70)	0.91 (0.73–1.13)
Acquisition of disease-causing genogroup	272 (2.13)	238 (2.07)	1.03 (0.79–1.34)



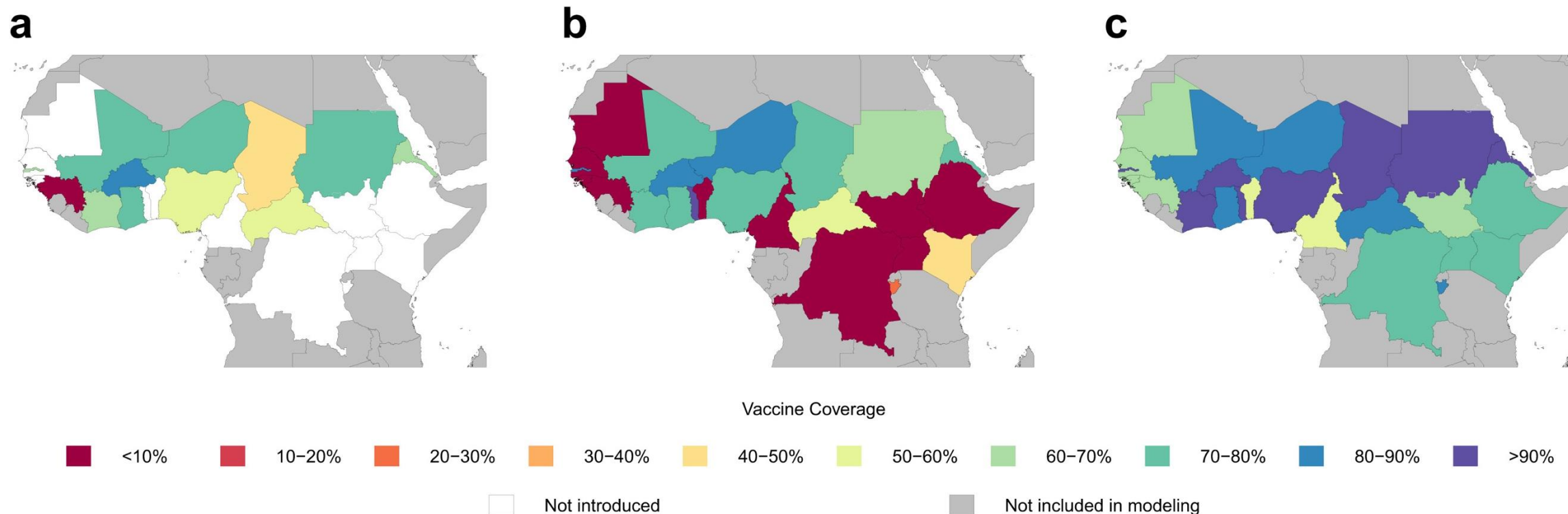
Helen Marshall

Marshall, H. S. & Maiden, M. C. J. (2021). Impact of a Meningococcal Protein-based Serogroup B Vaccine on Serogroup W Invasive Disease in Children. *Clinical Infectious Diseases*. **73**, E1669-E1671.

Protein/vesicle-based vaccines

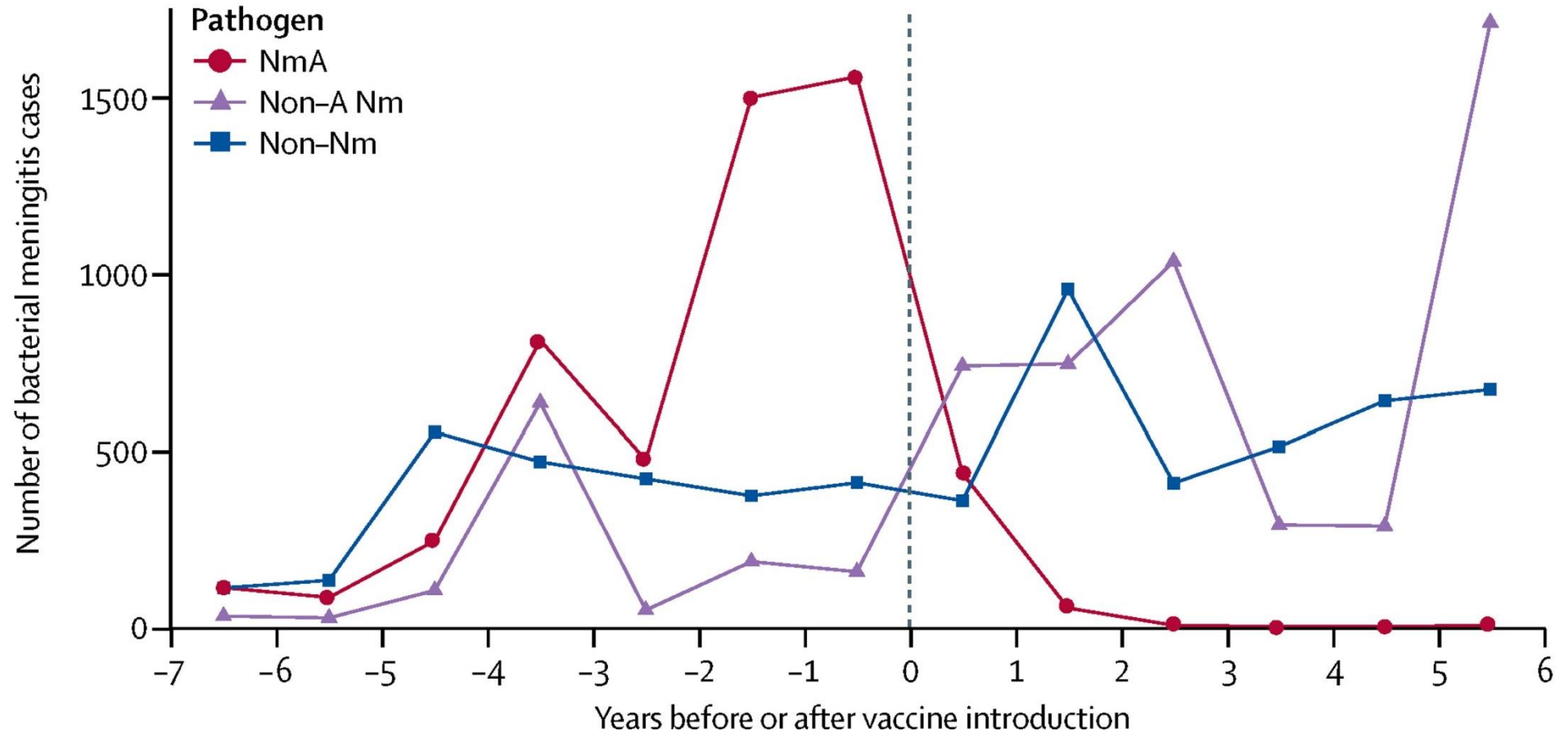


MenAfriVac[®] coverage estimates, year end 2021



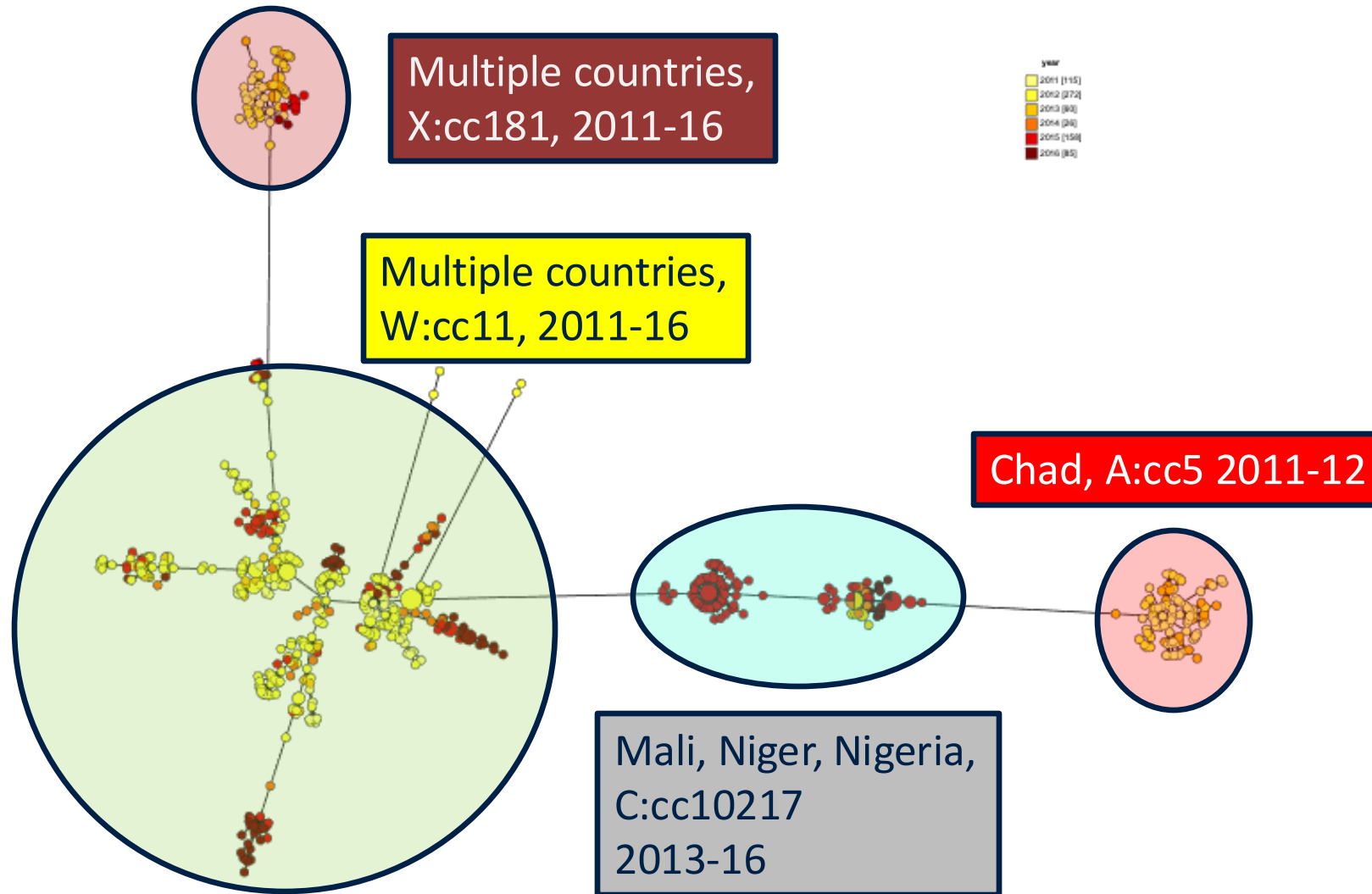
Bender, R.G. et al. (2023) Meningococcal A conjugate vaccine coverage in the meningitis belt of Africa from 2010 to 2021: A modelling study. *eClinical Medicine* 101797. DOI: 10.1016/j.eclinm.2022.101797

Laboratory confirmed meningitis cases in Africa



Trotter, C. L., Lingani, C., Fernandez, K., Cooper, L. V., Bitá, A., Tevi-Benissan, C., Ronveaux, O., Preziosi, M. P. & Stuart, J. M. (2017). Impact of MenAfriVac in nine countries of the African meningitis belt, 2010-15: an analysis of surveillance data. *Lancet Infectious Diseases* **17, 867-872.**

Genomes: African IMD Isolates 2011-2016



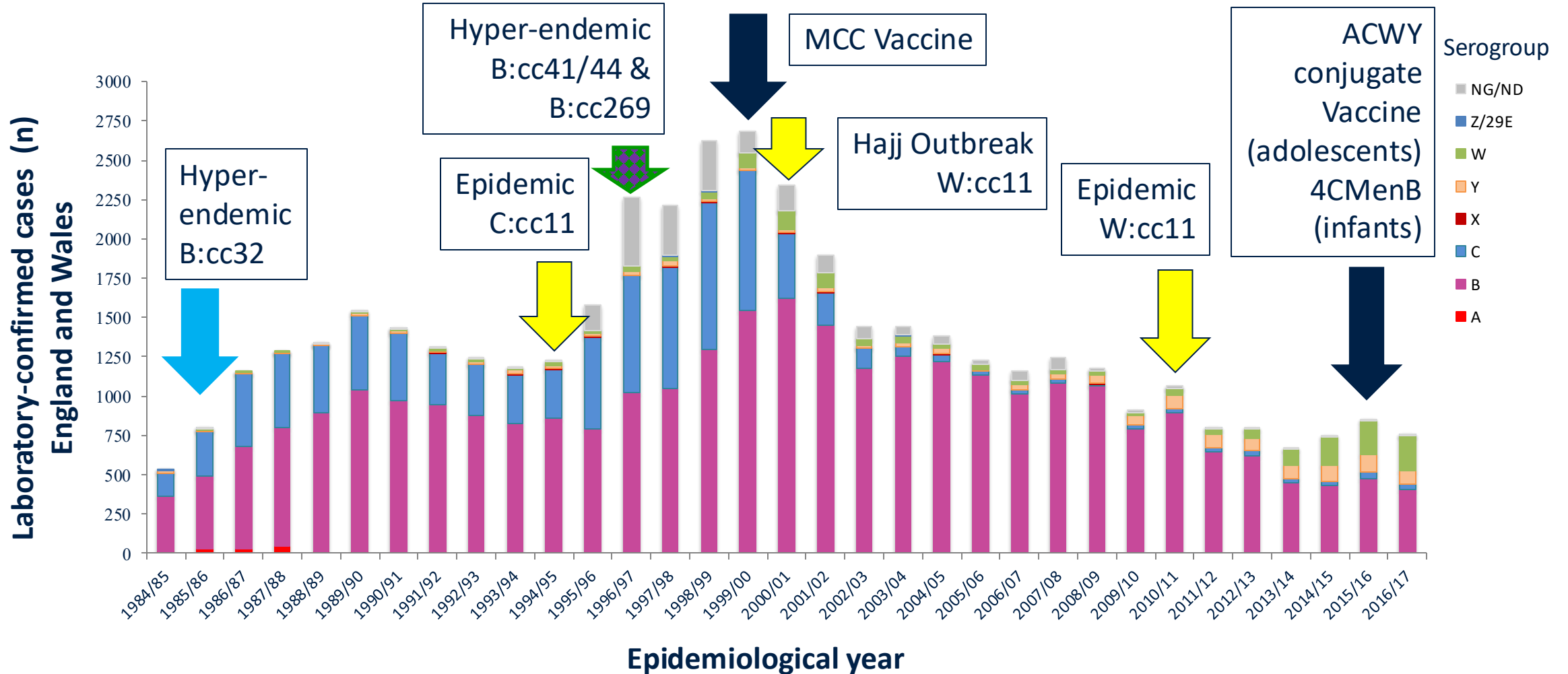
Data from:

Topaz, N., Caugant, D. A., Taha, M. K., Brynildsrud, O. B., Debech, N., Hong, E., Deghmane, A. E., Ouedraogo, R., Ousmane, S., Gamougame, K., Njanpop-Lafourcade, B. M., Diarra, S., Fox, L. M. & Wang, X. (2019). Phylogenetic relationships and regional spread of meningococcal strains in the meningitis belt, 2011-2016. *EBioMedicine* **41**, 488-496.

Analysis:

PubMLST, GrapeTree, cgMLST 1.0

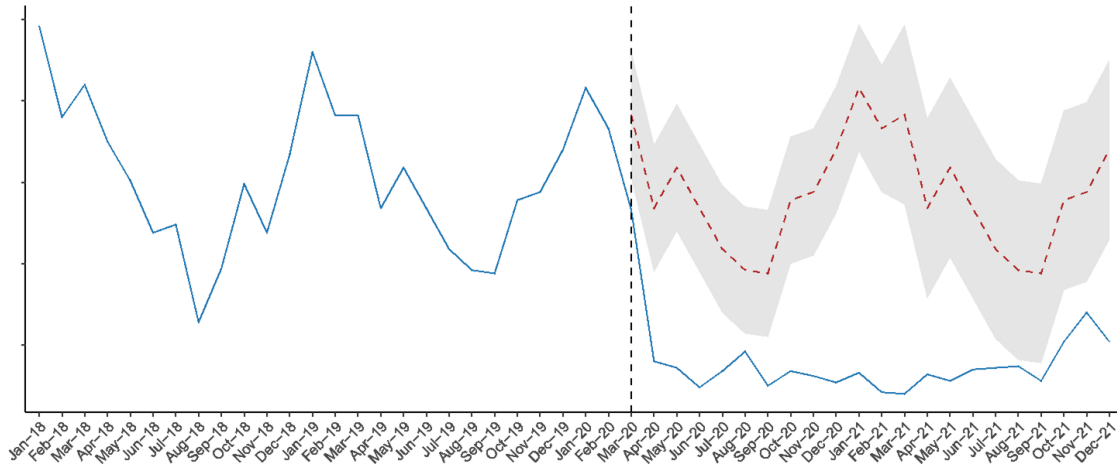
UK IMD epidemics and vaccination



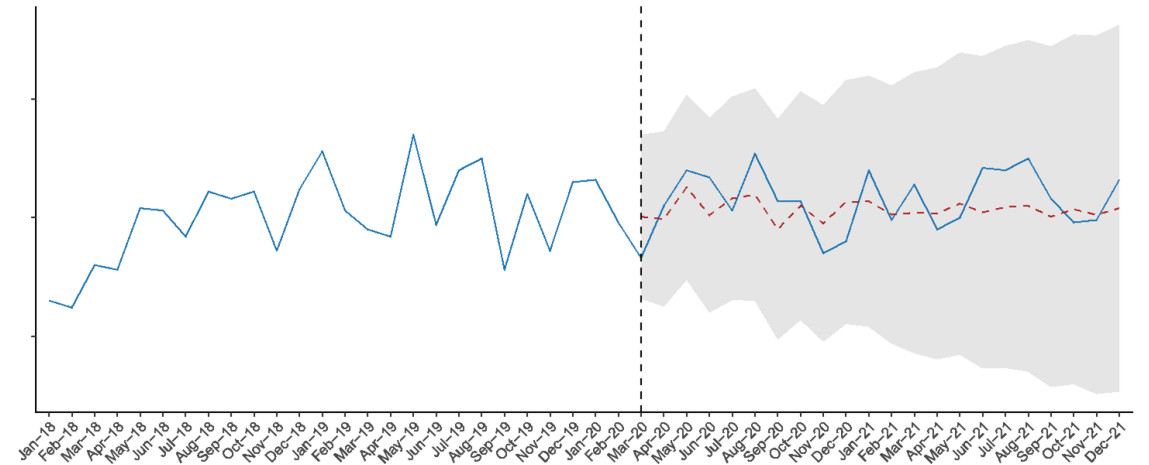
MacLennan, J. M., Rodrigues, C. M. C., Bratcher, H. B., Lekshmi, A., Finn, A., Oliver, J., Wootton, M., Ray, S., Cameron, C., Smith, A., Heath, P. T., Bartolf, A., Nolan, T., Hughes, S., Varghese, A., Snape, M. D., Sewell, R., Cunningham, R., Stolton, A., Kay, C., Palmer, K., Baxter, D., Suggitt, D., Zipitis, C. S., Pemberton, N., Jolley, K. A., Bray, J. E., Harrison, O. B., Ladhani, S. N., Pollard, A. J., Borrow, R., Gray, S. J., Trotter, C. & Maiden, M. C. J. (2021). Meningococcal carriage in periods of high and low invasive meningococcal disease incidence in the UK: comparison of UKMenCar1-4 cross-sectional survey results. *Lancet Infect Dis.* **21**, 677-687

COVID-19 and beyond: meningococcal and GBS disease

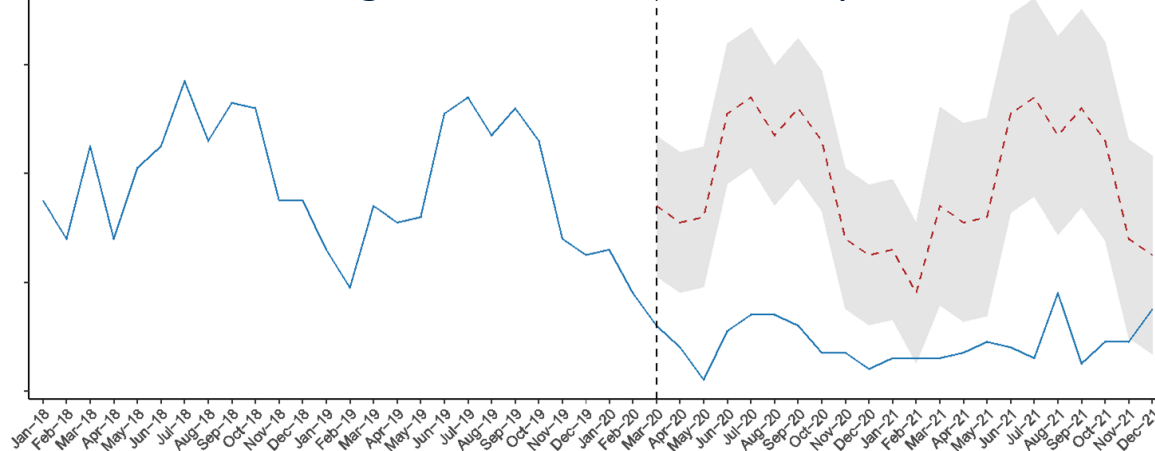
Neisseria meningitidis – Northern Hemisphere



Streptococcus agalactiae – Northern Hemisphere



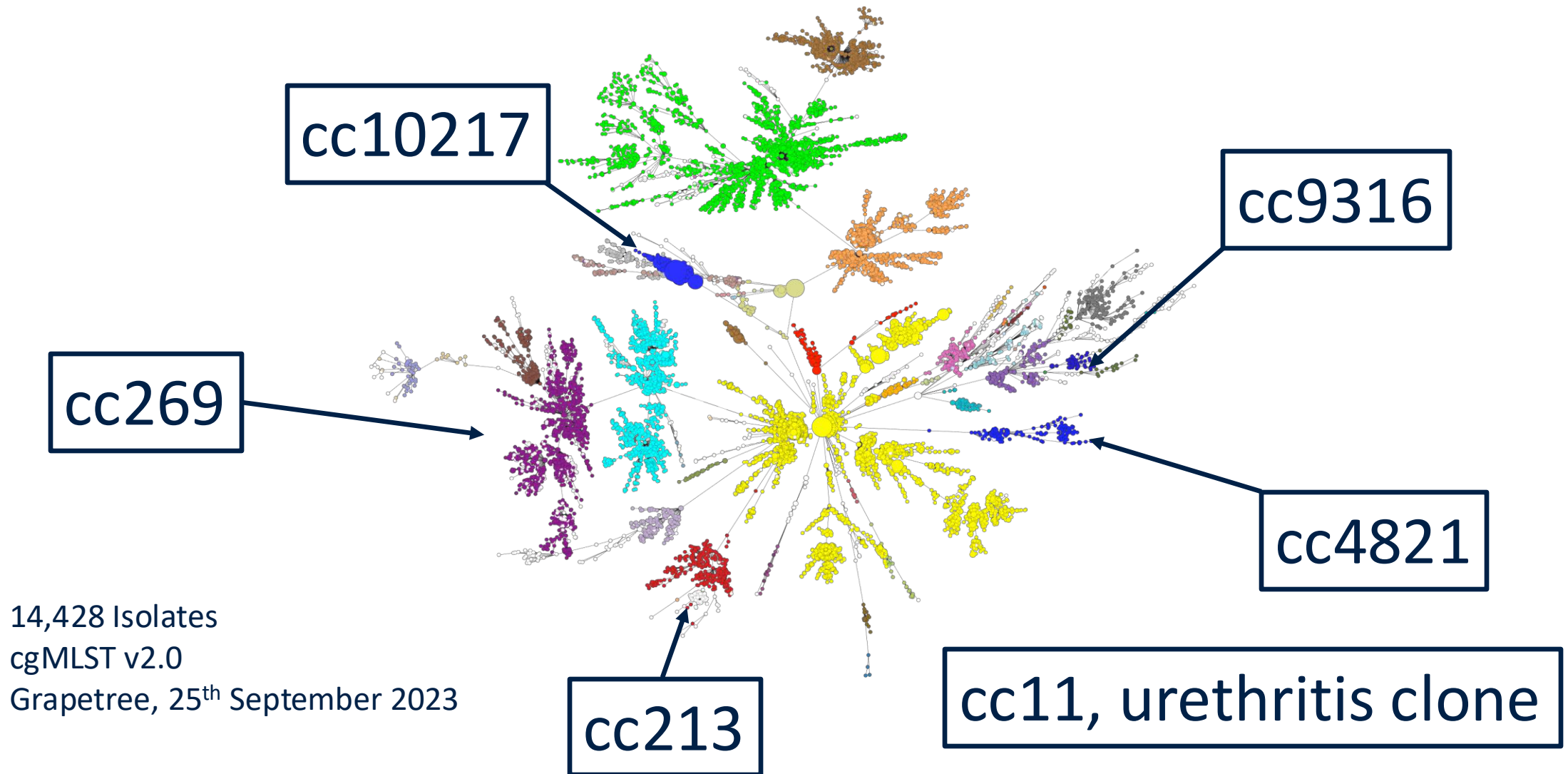
Neisseria meningitidis – Southern Hemisphere



<https://www.medrxiv.org/content/10.1101/2022.12.16.22283251v1>

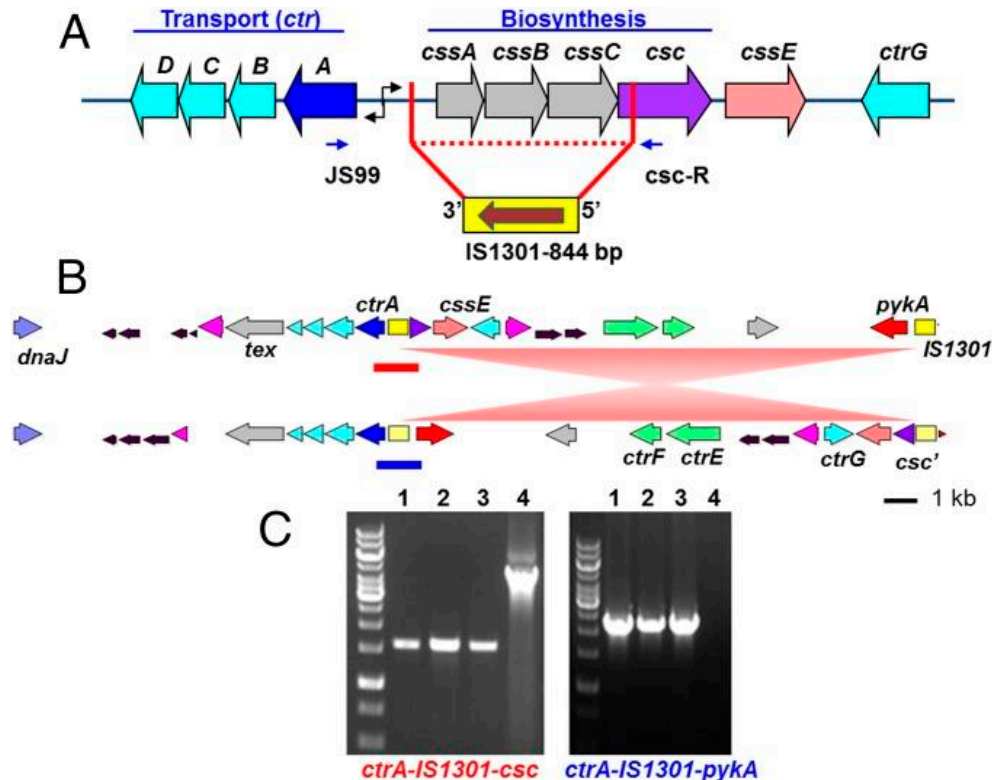


New meningococcal variants of concern

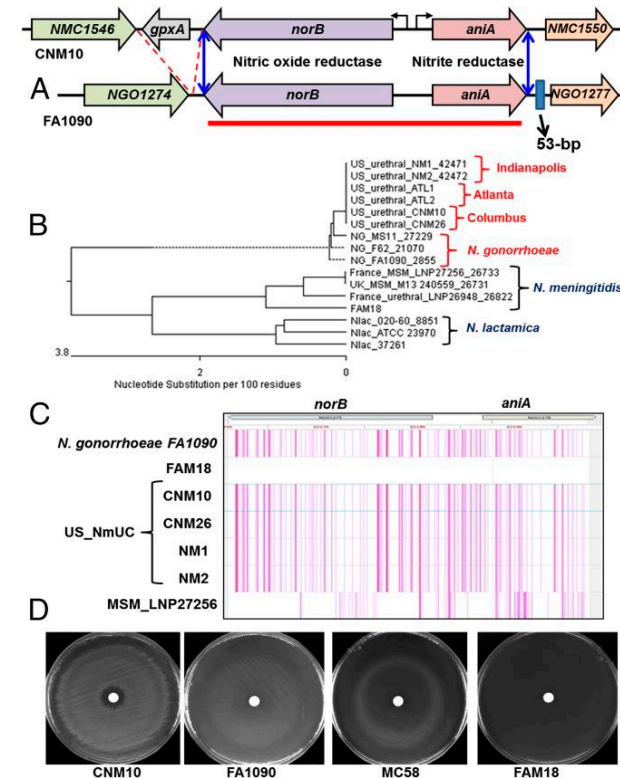


cc11 niche transition: a new gonococcus?

Inactivation of the capsule locus



Acquisition of *norB-aniA* gene cassette



Tzeng, Y. L., Bazan, J. A., Turner, A. N., Wang, X., Retchless, A. C., Read, T. D., Toh, E., Nelson, D. E., Del Rio, C. & Stephens, D. S. (2017). Emergence of a new *Neisseria meningitidis* clonal complex 11 lineage 11.2 clade as an effective urogenital pathogen. *Proc Natl Acad Sci U S A.* **114**, 4237-4242.



<https://www.who.int/initiatives/defeating-meningitis-by-2030>

... For how long?